A Coupled Tensor Double-Factor Method for Hyperspectral and Multispectral Image Fusion

Ting Xu, Ting-Zhu Huang, *Member, IEEE*, Liang-Jian Deng, *Senior Member, IEEE*, Jin-Liang Xiao, *Graduate Student Member, IEEE*, Clifford Broni-Bediako, *Member, IEEE*, Junshi Xia, *Senior Member, IEEE*, and Naoto Yokoya, *Member, IEEE*

***Abstract*— Hyperspectral image (HSI) and multispectral image (MSI) fusion, denoted as HSI-MSI fusion, involves merging a pair of HSI and MSI to generate a high spatial resolution HSI (HR-HSI). The primary challenge in HSI-MSI fusion is to find the best way to extract 1-D spectral features and 2-D spatial features from HSI and MSI and harmoniously combine them. In recent times, coupled tensor decomposition (CTD)-based methods have shown promising performance in the fusion task. However, the tensor decompositions (TDs) used by these CTD-based methods face difficulties in extracting complex features and capturing 2-D spatial features, resulting in suboptimal fusion results. To address these issues, we introduce a novel method called coupled tensor double-factor (CTDF) decomposition. Specifically, we propose a tensor double-factor (TDF) decomposition, representing a third-order HR-HSI as a fourth-order spatial factor and a third-order spectral factor, connected through a tensor contraction. Compared to other TDs, the TDF has better feature extraction capability since it has a higher order factor than that of HR-HSI, whereas the other TDs only have the same order factor as the HR-HSI. Moreover, the TDF can extract 2-D spatial features using the fourth-order spatial factor. We apply the TDF to the HSI-MSI fusion problem and formulate the CTDF model. Furthermore, we design an algorithm based on proximal alternating minimization (PAM) to solve this model and provide insights into its computational complexity and convergence analysis. The simulated and real experiments validate the effectiveness and efficiency of the proposed CTDF method. The code is available at https://github.com/tingxu113/CTDF.**

Manuscript received 3 January 2024; revised 5 March 2024 and 26 March 2024; accepted 26 March 2024. Date of publication 15 April 2024; date of current version 26 April 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2020YFA0714001, in part by the National Science Foundation of China under Grant 12171072 and Grant 12271083, in part by the Natural Science Foundation of Sichuan Province under Grant 2022NSFSC0501, and in part by the Key Projects of Applied Basic Research in Sichuan Province under Grant 2020YJ0216. *(Corresponding authors: Ting-Zhu Huang; Liang-Jian Deng.)*

Ting Xu is with the School of Mathematical Sciences, University of Electronic Science and Technology of China, Chengdu 611731, China, and also with the RIKEN Center for Advanced Intelligence Project, Tokyo 103-0027, Japan (e-mail: tingxu113@163.com).

Ting-Zhu Huang, Liang-Jian Deng, and Jin-Liang Xiao are with the School of Mathematical Sciences, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: tingzhuhuang@126.com; liangjian.deng@uestc.edu.cn; jinliang_xiao@163.com).

Clifford Broni-Bediako and Junshi Xia are with the RIKEN Center for Advanced Intelligence Project, Tokyo 103-0027, Japan (e-mail: clifford.broni-bediako@riken.jp; junshi.xia@riken.jp).

Naoto Yokoya is with the Department of Complexity Science and Engineering, Graduate School of Frontier Sciences, The University of Tokyo, Chiba 277-8561, Japan, and also with the RIKEN Center for Advanced Intelligence Project, Tokyo 103-0027, Japan (e-mail: yokoya@k.u-tokyo.ac.jp).

Digital Object Identifier 10.1109/TGRS.2024.3389016

***Index Terms*— Image fusion, proximal alternating minimization (PAM), tensor decomposition (TD), tensor double-factor (TDF) decomposition.**

NOMENCLATURE

Notation	Explanation
$\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times \cdots \times n_d}$	dth-order tensor with the size of $n_1 \times n_2 \times \cdots \times n_d$.
$\mathcal{X}(i_1, i_2, \cdots, i_d)$	Element of $\mathcal{X}$ indexed by $(i_1, i_2, \cdots, i_d)$.
$\|\|\mathcal{X}\|\|_F$	Frobenius norm $\|\|\mathcal{X}\|\|_F = \sqrt{\sum_{i_1 i_2 \cdots i_d} \mathcal{X}^2(i_1, i_2, \cdots i_d)}$.
$\mathbf{X} \in \mathbb{R}^{n_1 \times n_2}$	Matrix with the size of $n_1 \times n_2$.
$\mathbf{X}(i_1, i_2)$	Element of $\mathbf{X}$ indexed by (i_1, i_2).
$\mathbf{X}^T$	Transpose matrix of $\mathbf{X}$.
$\mathbf{x} \in \mathbb{R}^{n_1}$	Vector with the size of $1 \times n_1$.
$\mathbf{x}(1:j)$	First j elements in $\mathbf{x}$.
$\mathbf{I}$	Identity matrix.
$\text{vec}(\cdot)$	Vectorization operator.
$\otimes$	Kronecker product.

I. INTRODUCTION

HYPERSPECTRAL imaging sensors (optical imaging systems) offer advanced capabilities for capturing images across a wide range of spectra, spanning from the visible band to the shortwave infrared band. This comprehensive spectral feature acquisition empowers hyperspectral images (HSIs) to be used for precise material identification and differentiation. HSIs have found successful applications in various domains, including soil investigation [1], [2] and remote sensing [3], [4], [5], [6], [7], [8], [9]. Nevertheless, optical imaging systems inherently face tradeoffs among spatial resolution, spectral resolution, and signal-to-noise ratio due to limitations in solar irradiance. In particular, HSIs have an insufficient spatial resolution, which inevitably impacts their applications [10], [11], [12], [13], [14]. Developing new optical imaging systems to enhance the spatial resolution of HSIs has significant cost implications. Fortunately, fusing an HSI and a multispectral image (MSI) also denoted as HSI-MSI fusion, provides a practical and cost-effective solution for enhancing the spatial resolution of HSI. HSI-MSI fusion involves merging the 1-D spectral features with the 2-D spatial

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:24:23 UTC from IEEE Xplore. Restrictions apply.

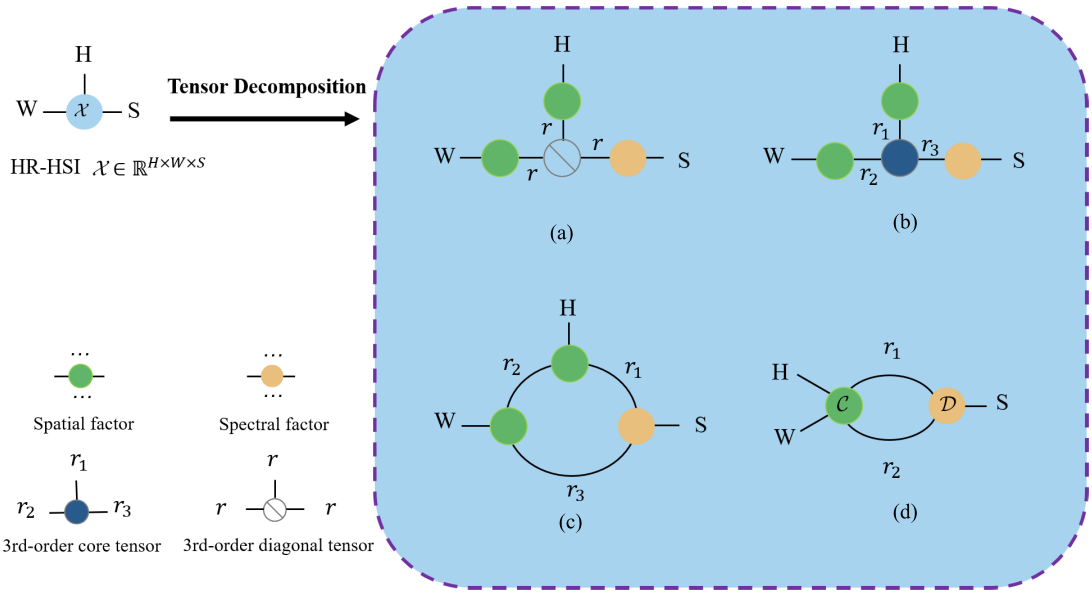


Fig. 1. Graphical representation of HR-HSI $\mathcal{X} \in \mathbb{R}^{H\times W\times S}$ using different TDs, i.e., (a) CP decomposition with CP-rank r, (b) Tucker decomposition with Tucker-rank $[r_1, r_2, r_3]$, (c) TR decomposition with TR-rank $[r_1, r_2, r_3]$, and (d) TDF decomposition with TDF-rank $[r_1, r_2]$, fourth-order spatial factor $\mathcal{C}$ and third-order spectral factor $\mathcal{D}$. Therein, the spatial factor denotes the tensor that contains at least one spatial mode (height mode, width mode), the spectral factor represents the tensor containing the spectral mode, and the core tensor and the diagonal tensor do not contain the spatial and spectral modes.

features of the HSI and the MSI to generate a high spatial resolution HSI (HR-HSI) [3].

Recent advancements in coupled tensor decomposition (CTD)-based methods [15], [16], [17] have demonstrated promising results in HSI-MSI fusion by leveraging the 3-D structure in HSI and MSI through various tensor decompositions (TDs). TD is a pivotal tool for feature extraction, adept at capturing multilinear and multifaceted structures in large-scale high-order data. Expanding upon canonical polyadic (CP) decomposition, Kanatsoulis et al. [15] introduced a coupled CP decomposition method for HSI-MSI fusion. This method assumes that the HR-HSI possesses a uniform low-rank property across its three dimensions (i.e., height, width, and spectral modes). However, recent observations suggest that the HR-HSI exhibits distinct low-rank characteristics along the three modes [18]. Li et al. [16] introduced a coupled Tucker decomposition model for HSI-MSI fusion, representing the HR-HSI as a core tensor multiplied by dictionary matrices along the three modes. Notably, the core tensor does not contain both spatial and spectral features, rendering it independent of the spatial and spectral degradation processes from the HR-HSI to the HSI and the HR-HSI to the MSI, respectively. More recently, He et al. [17] proposed the coupled tensor ring factorization (CTRF) for HSI-MSI fusion. CTRF combines the simplicity of coupled CP with the flexibility of coupled Tucker factorization to introduce spectral nuclear norm regularization to explore the global spectral low-rank property. This enables the CTRF to produce good fusion results.

It is well-known that higher order tensors generally possess the potential to capture more complex features in the data than lower order tensors. For instance, matrices describe the structure of data more effectively than vectors. Therefore, in the design of the TD format, introducing a higher order tensor than the order of the data may be advantageous for enhancing the feature extraction capability of TD. However, as depicted in Fig. 1(a)–(c), the CP, the Tucker, and the TR decomposition that have been employed in the existing CTD-based methods do not introduce factors of an order higher than the HR-HSI's third order. Consequently, the feature extraction capability of these three TDs may be limited. Furthermore, these TDs partition the 2-D spatial features in the HR-HSI into two parts. This often leads to the degradation of spatial features. In summary, the existing CTD-based methods face two challenges: 1) limited feature extraction capability and 2) inability to extract 2-D spatial features. These challenges may ultimately lead to suboptimal fusion results.

To address these challenges, we introduce the coupled tensor double-factor (CTDF) decomposition method for HSI-MSI fusion. We propose a tensor double-factor (TDF) decomposition, as illustrated in Fig. 1(d). This decomposition represents the third-order HR-HSI as a fourth-order spatial factor and a third-order spectral factor connected through tensor contraction. The TDF decomposition offers two distinct advantages over other TDs: 1) it showcases superior feature extraction capability compared to the TR decomposition, primarily attributed to the introduction of a fourth-order factor, as elaborated in Section V-B1; and 2) it can extract 2-D spatial features using the fourth-order spatial factor (elaborated in Section V-B2). The core concept of the proposed CTDF method is as follows: first, we establish a TDF degradation model that consists of two components (i.e., the HR-HSI to MSI spectral degradation and the HR-HSI to HSI spatial degradation). In this model, the HR-HSI is degraded to MSI through the downsampling of the third-order spectral factor (i.e., the HR-HSI to MSI spectral degradation), and it is degraded to HSI through the downsampling of the fouth-order spatial factor (i.e., the HR-HSI to HSI spatial degradation), as depicted in Fig. 2. Then, we formulate the CTDF model

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:24:23 UTC from IEEE Xplore. Restrictions apply.

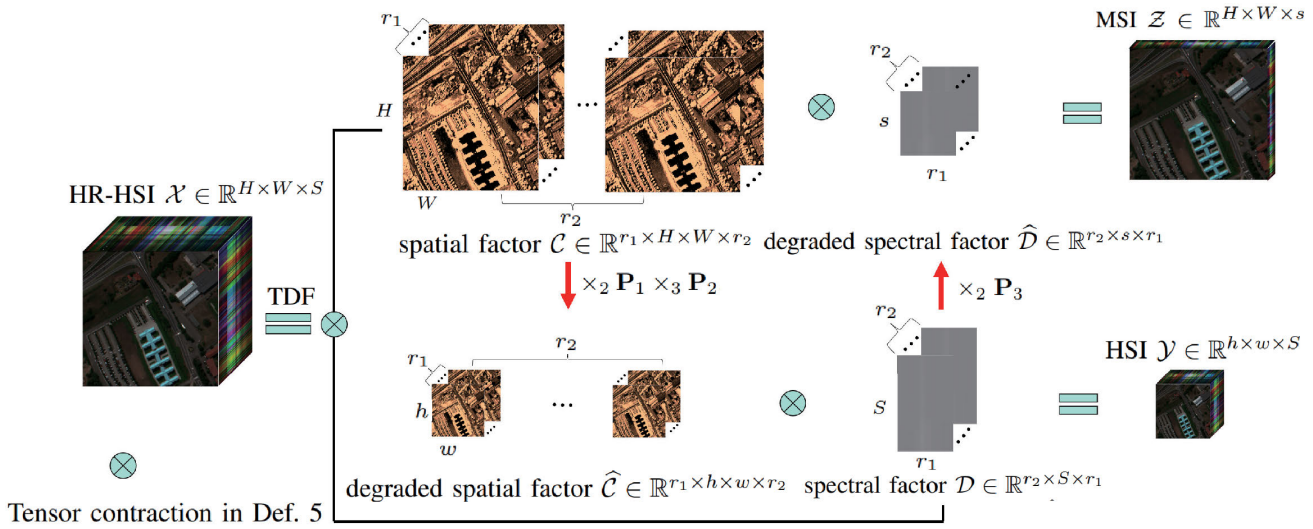


Fig. 2. TDF degradation model from the HR-HSI to both MSI and HSI. (Top) From the HR-HSI $\mathcal{X} \in \mathbb{R}^{H\times W\times S}$ to MSI $\mathcal{Z} \in \mathbb{R}^{H\times W\times s}$ spectral degradation. (Bottom) From the HR-HSI $\mathcal{X} \in \mathbb{R}^{H\times W\times S}$ to HSI $\mathcal{Y} \in \mathbb{R}^{h\times w\times S}$ spatial degradation.

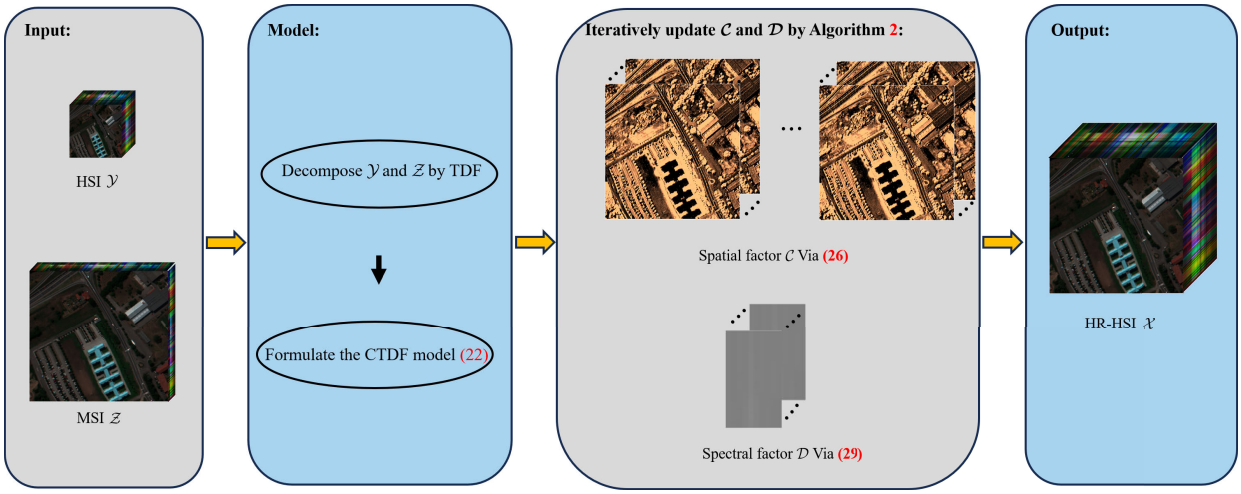


Fig. 3. Flowchart of the proposed CTDF method for HSI-MSI fusion.

and design an algorithm based on proximal alternating minimization (PAM) to iteratively update the spatial and spectral factors. The final step involves reconstructing the HR-HSI from the optimized spatial and spectral factors using the TDF decomposition, as shown in Fig. 3. The main contributions of this article can be summarized as follows.

1) We introduce a novel TDF decomposition and demonstrate its superior feature extraction capability over the TR using a practical example. Compared to other TDs, the TDF can capture 2-D spatial features by the fourth-order spatial factor.
2) Building upon the TDF decomposition, we develop a degradation model that simulates the transformation from HR-HSI to both MSI and HSI. This forms the basis for formulating the CTDF model for the HSI-MSI fusion problem.
3) We design a PAM-based algorithm to solve the CTDF model and provide detailed insights into its computational complexity and convergence analysis. Simulated experiments conducted on four publicly available datasets and real experiments on a pair of real datasets show that the proposed CTDF method outperforms state-of-the-art HSI-MSI fusion methods.

We organize the remaining sections of this article as follows. Section II offers a comprehensive summary and discussion of various approaches to HSI-MSI fusion. In Section III, we establish the necessary foundations by presenting preliminaries. Section IV introduces the TDF decomposition, while a detailed overview of the CTDF methodology is provided in Section V. Section VI presents the experiments performed on HSI-MSI fusion, followed by exploring the parameters of the CTDF method and analyzing the TDF in Section VII. Finally, in Section VIII, we draw our conclusion.

II. Related Work

Recently, there has been a noticeable upsurge of interest among researchers in addressing the challenges of HSI-MSI fusion, leading to the development of diverse approaches to tackle this problem. We classify these HSI-MSI

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:24:23 UTC from IEEE Xplore. Restrictions apply.

fusion approaches into three main categories: deep learning (DL)-based, matrix factorization (MF)-based, and TD-based methods.

A. DL-Based Approaches

The existing DL-based methods, especially the data-driven convolutional neural network (CNN)-based methods, have shown superior fusion accuracy [19], [20]. CNN-based methods are designed to learn the mapping function from HSI and MSI to the target HR-HSI [21], [22], [23], [24], [25]. Dian et al. [26] introduced an approach that initializes the high-resolution HSI using a Sylvester equation and then employs a deep CNN to learn the mapping function from the initialized image to a reference image. Shen et al. [27] integrated PixelShuffle and attention modules to extract fusion features. These ideas have been further developed in subsequent works [28], [29]. Palsson et al. [30] introduced an iterative refinement unit to enhance fusion accuracy by reducing reconstruction errors. Moreover, several CNN-based approaches [31], [32], [33] have leveraged imaging models for HSI-MSI fusion. For example, Dong et al. [33] unfolded an iterative optimization framework into an end-to-end network. Recently, the diffusion model has received increasing attention in the field of HSI-MSI fusion due to its powerful generation capability [34], [35], [36], [37], [38]. Cao et al. [35] introduced a supervised diffusion model with two conditional modulation modules for the HSI-MSI fusion task. These supervised methods rely on the "invariance among scale" hypothesis. To overcome this drawback, unsupervised DL methods are introduced for HSI-MSI fusion. Zhang et al. [39] proposed an adaptation learning framework for HSI-MSI fusion. Li et al. [40] introduced a model-guided coarse-to-fine fusion network. Gao et al. [41] utilized attention-embedded degradation learning for an unsupervised strategy. Hong et al. [5] proposed decoupled-and-coupled structures. While these methods have achieved excellent results, they still exhibit three inherent drawbacks. First, DL-based methods rely on extensive training data for optimal learning, which can be challenging to obtain in practice, especially when acquiring image data consisting of HSI, MSI, and HR-HSI of the same scene for supervised methods. Second, the generalization ability of DL-based methods is influenced by variations in data characteristics, making them less effective when the spectral ranges, the number of spectral bands, or the imaging models of the training data differ from those of the test data. Third, the issue of interpretability remains a notable concern. Extensive research has been conducted to address these challenges [42], [43], [44]. Dian et al. [44] proposed an advanced method called the zero-shot learning (ZSL) method, which effectively addresses the first two problems. However, the issue of interpretability still poses a challenge in the DL-based methods community.

B. MF-Based Approaches

Recently, MF-based methods have demonstrated significant effectiveness in HSI-MSI fusion. MF-based fusion methods can be broadly classified into uncoupled MF (UMF)-based methods and coupled MF (CMF)-based methods. The UMF-based methods do not alternatively optimize the spectral basis and coefficients of HR-HSI in the matrix. Some studies [45], [46], [47] argue that the spectral features and spatial features mainly lie in the HSI and MSI, respectively. They estimate the spectral basis only from the HSI and obtain the coefficients only from the MSI. Other researchers [48], [49], [50], [51], [52] also argue that the HSI contains spatial features and can contribute to the estimation of the coefficients. Here, the spectral basis is computed from HSI only, and then the coefficients are calculated from both HSI and MSI. Recently, Xue et al. [53] introduced a novel UMF-based method, S^4-LRR, which diverges from the aforementioned UMF-based methods. The S^4-LRR introduced structured sparse low-rank representation (SSLRR), which separately learns the spatial SSLRR from MSI and the spectral SSLRR from HSI. Then, the acquired spatial-spectral SSLRR is integrated into the MSI-HSI fusion framework to regulate the spatial-spectral prior to HR-HSI. The CMF-based methods follow a different approach, involving the alternate updating of spectral basis and coefficients. A notable representative work of this approach is the coupled nonnegative MF (CNMF) proposed by Yokoya et al. [54]. Subsequently, several other CMF-based methods [55], [56], [57] have been proposed, showing satisfactory fusion performance. Nevertheless, both UMF-based and CMF-based approaches grapple with noteworthy challenges. Specifically, the natural representation of HR-HSI as a third-order tensor clashes with the unfolding operator that is imposed by these methods, which transforms the height-width-spectrum structure into a height $\times$ width-spectrum matrix. This transformation disrupts the inherent 3-D structure of HR-HSI.

C. TD-Based Approaches

To address the inherent limitation (i.e., the disruption of the 3-D structure of the HR-HSI) of the MF-based methods, TD-based techniques have emerged as a promising alternative in preserving the original height-width-spectrum structure of the HR-HSI. These techniques can be categorized into two families: uncoupled TD (UTD)-based methods and CTD-based methods. The UTD-based methods [58], [59], [60], [61] establish the HSI-MSI fusion model in tensor format and leverage the properties of HR-HSI through TD techniques. For instance, Dian et al. [60] proposed a low tensor-train rank regularized HSI–MSI fusion method, called LTTR, which makes use of the nonlocal spatial–spectral similarities of the HR-HSI. More recently, Xu et al. [59] proposed an IR-TenSR HSI-MSI fusion method, which introduced a tensor subspace to represent the HR-HSI, thereby characterizing its global spectral-spatial low-rank property. These methods cannot fully exploit the intrinsic structure in HSI and MSI since they do not explicitly represent them in TD formats, thus, inevitably affecting the fusion accuracy. In contrast, the CTD-based methods [15], [16], [17], [62], [63], [64] utilize TDs to represent the HR-HSI. These methods explicitly represent the HSI and the MSI in their corresponding TD formats based on the degradation model, thereby establishing the fidelity terms. To further improve fusion accuracy, the CTD-based methods

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:24:23 UTC from IEEE Xplore. Restrictions apply.

usually integrate regularization terms into the HSI-MSI fusion model. Also, the factors of TD are updated alternately. Noteworthy methods in this category include the coupled CP-based method by Kanatsoulis et al. [15], the nonlocal coupled CP-based method by Xu et al. [62], and the coupled Tucker decomposition by Li et al. [16]. Building on Li et al. [16], Xu et al. [63] introduced total variation (TV) regularization to enhance fusion performance. Recent contributions, such as the coupled TR-based method with factor-smoothed regularization by Chen et al. [64] and the coupled TR-based method with a global spectral low-rank property by He et al. [17], have further enriched the repertoire of CTD-based approaches. However, these methods usually produce suboptimal fusion results due to the drawbacks mentioned in Section I.

III. Preliminaries

In this article, we adopt the following notation conventions: vectors, matrices, and tensors are denoted as boldface lowercase letters, e.g., $\mathbf{x}$, boldface capital letters, e.g., $\mathbf{X}$, and calligraphic letters, e.g., $\mathcal{X}$, respectively. Nomenclature provides a summary of basic tensor notations along with their corresponding explanations. In addition, the following are definitions related to this work.

Definition 1 (Diagonal Tensor [65]): Suppose that $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots I_d}$ is a dth-order tensor. $\mathcal{X}$ is a diagonal tensor if $\mathcal{X}(i_1, i_2, \ldots, i_d) \neq 0$ only if $i_1 = i_2 = \cdots = i_d$.

Definition 2 (k-Mode Product [65]): Suppose that $\mathcal{C} \in \mathbb{R}^{I_1 \times \cdots \times I_{k-1} \times I_k \times I_{k+1} \times \cdots \times I_M}$ and $\mathbf{D} \in \mathbb{R}^{J \times I_k}$ ($1 \leq k \leq M$, $k \in \mathbb{Z}$) are an Mth-order tensor and a matrix, respectively. The k-mode product between $\mathcal{C}$ and $\mathbf{D}$ is denoted by $\mathcal{C} \times {}_k\mathbf{D}$ and is an Mth-order tensor $\mathcal{G} \in \mathbb{R}^{I_1 \times \cdots \times I_{k-1} \times J \times I_{k+1} \times \cdots \times I_M}$. Element-wise, we have

$$\mathcal{G}(i_1, \ldots, i_{k-1}, j, i_{k+1}, \ldots, i_M) = \sum_{i_k=1}^{I_k} \mathcal{C}(i_1, \ldots, i_k, \ldots, i_M)\mathbf{D}(j, i_k). \quad (1)$$

Generally, given a collection of matrices $\mathbf{D}_m \in \mathbb{R}^{J_m \times I_m}$ ($m = 1, 2, \ldots, M$), we define the tensor $\mathcal{G} \in \mathbb{R}^{J_1 \times J_2 \times \cdots \times J_M}$ as

$$\mathcal{G} = \mathcal{C} \times_1 \mathbf{D}_1 \times_2 \mathbf{D}_2 \times \cdots \times_M \mathbf{D}_M. \quad (2)$$

Then, we have

$$\mathbf{g} = (\mathbf{D}_M \otimes \mathbf{D}_{M-1} \otimes \cdots \otimes \mathbf{D}_1)\mathbf{c} \quad (3)$$

where $\mathbf{g} = \text{vec}(\mathcal{G}) \in \mathbb{R}^{\prod_{m=1}^{M} J_M \times 1}$ and $\mathbf{c} = \text{vec}(\mathcal{C}) \in \mathbb{R}^{\prod_{m=1}^{M} I_m \times 1}$, $\text{vec}(\cdot)$ denotes the vectorization operator, and the symbol "$\otimes$" represents Kronecker product.

Definition 3 (Tensor Permutation [66]): Given an Mth-order tensor $\mathcal{C} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$. Assuming that vector $\mathbf{m} = [m_1, m_2, \ldots, m_M]$ is a reordering of vector $[1, 2, \ldots, M]$, then we define the vector $\mathbf{m}$-based permutation of tensor $\mathcal{C}$ as a tensor $\vec{\mathcal{C}}^{\mathbf{m}} \in \mathbb{R}^{I_{m_1} \times I_{m_2} \times \cdots \times I_{m_M}}$, whose elements satisfy

$$\vec{\mathcal{C}}^{\mathbf{m}}(i_{m_1}, i_{m_2}, \ldots, i_{m_M}) = \mathcal{C}(i_1, i_2, \ldots, i_M). \quad (4)$$

Definition 4 (Generalized Tensor k-Unfolding [66]): Let $\mathbf{m} = [m_1, m_2, \ldots, m_M]$ be a reordering of vector $[1, 2, \ldots, M]$, then for an Mth-order tensor $\mathcal{C} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$, the generalized k-unfolding ($0 \leq k \leq M, k \in \mathbb{Z}$) of $\mathcal{C}$ is a matrix, denoted by $\mathbf{C}_{[\mathbf{m};k]}$ of size $\prod_{i=1}^{k} I_{m_i} \times \prod_{j=k+1}^{M} I_{m_j}$, whose elements are defined by

$$\mathbf{C}_{[\mathbf{m};k]}\left(\overline{i_{m_1} i_{m_2}, \ldots, i_{m_k}}, \overline{i_{m_{k+1}} i_{m_{k+2}}, \ldots, i_{m_M}}\right) = \vec{\mathcal{C}}^{\mathbf{m}}(i_{m_1}, i_{m_2}, \ldots, i_{m_M}) \quad (5)$$

where the multiindices $\overline{i_{m_1} i_{m_2}, \ldots, i_{m_k}} = 1 + \sum_{s=1}^{k}(i_{m_s} - 1)\prod_{t=1}^{s-1} I_{m_t}$ and $\overline{i_{m_{k+1}} i_{m_{k+2}}, \ldots, i_{m_M}} = 1 + \sum_{s=k+1}^{M}(i_{m_s} - 1)\prod_{t=k+1}^{s-1} I_{m_t}$, $\vec{\mathcal{C}}^{\mathbf{m}}$ is the tensor permutation of $\mathcal{C}$. Its inverse operator is denoted as $\mathcal{C} = \text{Fold}_{[\mathbf{m};k]}(\mathbf{C}_{[\mathbf{m};k]})$.

Remark 1: When $\mathbf{m} = [1, 2, \ldots, M]$, $\mathbf{C}_{[\mathbf{m};k]}$ is indeed the tensor k-unfolding of $\mathcal{C}$ [67], denoted as $\mathbf{C}_{\langle k \rangle} \in \mathbb{R}^{\prod_{i=1}^{k} I_i \times \prod_{j=k+1}^{M} I_j}$.

Definition 5 (Generalized Tensor Contraction [66]): Given an Mth-order tensor $\mathcal{C} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$ and a Nth-order tensor $\mathcal{D} \in \mathbb{R}^{J_1 \times J_2 \times \cdots \times J_N}$ with k common modes ($1 \leq k \leq \min(M, N), k \in \mathbb{Z}$). Assume that two vectors $\mathbf{m} = [m_1, m_2, \ldots, m_M]$ and $\mathbf{n} = [n_1, n_2, \ldots, n_N]$, respectively, indicate the reordering of vectors $[1, 2, \ldots, M]$ and $[1, 2, \ldots, N]$, satisfying $I_{m_i} = J_{n_i}$ for $i = 1, 2, \ldots, k$, $m_{k+1} < m_{k+2} < \cdots < m_M$ and $n_{k+1} < n_{k+2} < \cdots < n_N$. Then, the generalized k-contraction between $\mathcal{C}$ and $\mathcal{D}$ along the k modes specifies an $(M + N - 2k)$th-order tensor, denoted as $\mathcal{C} \times {}^{\mathbf{n}(1:k)}_{\mathbf{m}(1:k)} \mathcal{D}$, is defined by

$$\mathcal{C} \times {}^{\mathbf{n}(1:k)}_{\mathbf{m}(1:k)} \mathcal{D} = \texttt{Fold}_{[[1,2,\ldots,M+N-2k];M-k]}\left(\mathbf{C}^T_{[\mathbf{m};k]}\mathbf{D}_{[\mathbf{n};k]}\right) \quad (6)$$

where $\mathbf{C}_{[\mathbf{m};k]}$ and $\mathbf{D}_{[\mathbf{n};k]}$ are the $\mathbf{m}$- and $\mathbf{n}$-based k-unfolding of tensor $\mathcal{C}$ and $\mathcal{D}$, respectively.

IV. Proposed TDF Decomposition

Here, we describe the proposed TDF decomposition. Section IV-A introduces the novel TDF decomposition along with its algebraic properties analysis. Section IV-B presents the proposed alternating least squares (ALSs)-based algorithm for the TDF decomposition.

A. TDF Decomposition

As illustrated in Fig. 1(d), for a third-order HR-HSI $\mathcal{X} \in \mathbb{R}^{H \times W \times S}$, the TDF decomposition represents it as a fourth-order spatial factor, denoted as $\mathcal{C} \in \mathbb{R}^{r_1 \times H \times W \times r_2}$, and a third-order spectral factor, denoted as $\mathcal{D} \in \mathbb{R}^{r_2 \times S \times r_1}$. The element-wise form of the TDF decomposition can be expressed as follows:

$$\mathcal{X}(i_1, i_2, i_3) = \sum_{s=1}^{r_1} \sum_{t=1}^{r_2} \mathcal{C}(s, i_1, i_2, t)\mathcal{D}(t, i_3, s). \quad (7)$$

Here, we define the TDF-rank as a vector $\mathbf{r} = [r_1, r_2]$. The TDF representation (7) can be expressed in tensor format using generalized tensor contraction (refer to Definition 5) as follows:

$$\mathcal{X} = \mathcal{C} \times {}^{1,3}_{4,1} \mathcal{D}. \quad (8)$$

In what follows, we will delve into the algebraic properties of the TDF decomposition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:24:23 UTC from IEEE Xplore. Restrictions apply.

Theorem 1: Assume that the TDF decomposition of tensor $\mathcal{X} \in \mathbb{R}^{H\times W\times S}$ is $\mathcal{X} = \mathcal{C} \times_{4,1}^{1,3} \mathcal{D}$. Then, the following equality holds:

$$\mathbf{X}_{\langle 2\rangle} = \mathbf{C}_{[\mathbf{m}_1;2]}^T \mathbf{D}_{[\mathbf{m}_2;2]} \tag{9}$$

where $\mathbf{m}_1 = [4, 1, 2, 3]$ and $\mathbf{m}_2 = [1, 3, 2]$.

Proof: As per Definition 5 and Remark 1, (9) can be derived. □

Theorem 2: Assume that $\mathcal{X} = \mathcal{C} \times_{4,1}^{1,3} \mathcal{D}$ with spatial factor $\mathcal{C} \in \mathbb{R}^{r_1\times H\times W\times r_2}$ and spectral factor $\mathcal{D} \in \mathbb{R}^{r_2\times S\times r_1}$, then

$$\text{Rank}(\mathbf{X}_{\langle 2\rangle}) \le r_2 r_1. \tag{10}$$

Proof: According to Theorem 1, we have

$$\begin{aligned}\text{Rank}(\mathbf{X}_{\langle 2\rangle}) &= \text{Rank}(\mathbf{C}_{[\mathbf{m}_1;2]}^T \mathbf{D}_{[\mathbf{m}_2;2]}) \\ &\le \min\{\text{Rank}(\mathbf{C}_{[\mathbf{m}_1;2]}^T), \text{Rank}(\mathbf{D}_{[\mathbf{m}_2;2]})\} \\ &\le \min\{\text{Rank}(\mathbf{D}_{[\mathbf{m}_2;2]})\} \\ &\le \min\{r_2 r_1, S\} \\ &\le r_2 r_1. \end{aligned} \tag{11}$$

□

B. ALS-Based Algorithm for the TDF Decomposition

This section presents the ALS-based algorithm that we developed to perform the TDF decomposition. Given a third-order tensor $\mathcal{X} \in \mathbb{R}^{H\times W\times S}$, our goal is to optimize factors $\mathcal{C}$ and $\mathcal{D}$ with a predefined TDF-rank $\mathbf{r}$, i.e.,

$$\arg\min_{\mathcal{C},\mathcal{D}} \left\| \mathcal{X} - \mathcal{C} \times_{4,1}^{1,3} \mathcal{D} \right\|_F. \tag{12}$$

Based on Theorem 1, the minimizing problem (12) can be optimized by solving the following two subproblems alternatively:

$$\arg\min_{\mathbf{C}_{[\mathbf{m}_1;2]}} \left\| \mathbf{X}_{\langle 2\rangle} - \mathbf{C}_{[\mathbf{m}_1;2]}^T \mathbf{D}_{[\mathbf{m}_2;2]}^k \right\|_F \tag{13}$$

and

$$\arg\min_{\mathbf{D}_{[\mathbf{m}_2;2]}} \left\| \mathbf{X}_{\langle 2\rangle} - \left(\mathbf{C}_{[\mathbf{m}_1;2]}^{k+1}\right)^T \mathbf{D}_{[\mathbf{m}_2;2]} \right\|_F \tag{14}$$

where k is the number of iterations, $\mathbf{m}_1 = [4, 1, 2, 3]$ and $\mathbf{m}_2 = [1, 3, 2]$ are consistent with (9).

Then, the TDF decomposition is performed by ALS optimization, which is called the TDF-ALS. The detailed procedure of the TDF-ALS is shown in Algorithm 1.

Algorithm 1 The Alternating Least Squares Algorithm for TDF Decomposition (TDF-ALS)

Input: A 3rd-order tensor $\mathcal{X} \in \mathbb{R}^{H\times W\times S}$, the predefined TDF-rank $\mathbf{r}$, the maximum number of iterations k_{mit}, the threshold ϵ, $\mathbf{m}_1 = [4, 1, 2, 3]$, and $\mathbf{m}_2 = [1, 3, 2]$.

Output: TDF factors $\mathcal{C}$ and $\mathcal{D}$.

1: Initialize $k = 0$, $\mathcal{C}^0 \in \mathbb{R}^{r_1\times H\times W\times r_2}$ and $\mathcal{D}^0 \in \mathbb{R}^{r_2\times S\times r_1}$ as random tensors from Gaussian distribution.
2: **while** $k < k_{mit}$ and RE $> \epsilon$ **do**
3: Update $\mathbf{C}_{[\mathbf{m}_1;2]}^{k+1}$ by using (13).
4: $\mathcal{C}^{k+1} \leftarrow \text{Fold}_{[\mathbf{m}_1;2]}(\mathbf{C}_{[\mathbf{m}_1;2]}^{k+1})$.
5: Update $\mathbf{D}_{[\mathbf{m}_2;2]}^{k+1}$ by using (14).
6: $\mathcal{D}^{k+1} \leftarrow \text{Fold}_{[\mathbf{m}_2;2]}(\mathbf{D}_{[\mathbf{m}_2;2]}^{k+1})$.
7: Compute $\left\| \mathcal{X} - \mathcal{C}^{k+1} \times_{4,1}^{1,3} \mathcal{D}^{k+1} \right\|_F / \|\mathcal{X}\|_F$ as relative error (RE).
8: Update $k = k + 1$.
9: **end while**

V. Proposed Coupled TDF (CTDF) Decomposition Method for HSI-MSI Fusion

This section presents the proposed CTDF decomposition method for HSI-MSI fusion. Section V-A provides HSI-MSI fusion problem formulation. The motivation for using TDF in HSI-MSI fusion is presented in Section V-B. Section V-C introduces the proposed CTDF model and PAM-based algorithm. The computational complexity and the convergence proof of the PAM-based algorithm are provided in Sections V-D and V-E, respectively.

A. HSI-MSI Fusion Problem Formulation

In this article, the third-order tensors indexed of HR-HSI, HSI, and MSI of the same scene are considered as three exploratory variables (h, w, and s), where h, w, and s denote the indices for the height, width, and spectrum modes, respectively. The target HR-HSI is denoted as $\mathcal{X} \in \mathbb{R}^{H\times W\times S}$, with H, W, and S representing the dimensions of the first mode (height), second mode (width), and third mode (spectrum), respectively. The tensor $\mathcal{Y} \in \mathbb{R}^{h\times w\times S}$ denotes the HSI, which experiences spatial degradation of $\mathcal{X}$, thus, $H > h$ and $W > w$. The tensor $\mathcal{Z} \in \mathbb{R}^{H\times W\times s}$ represents the MSI, suffering spectral degradation of $\mathcal{X}$, thus, $S > s$.

The objective of HSI-MSI fusion is to generate the $\mathcal{X}$ from the $\mathcal{Y}$ and the $\mathcal{Z}$. The core essence of HSI-MSI fusion involves combining the 1-D spectral features and the 2-D spatial features from both $\mathcal{Y}$ and $\mathcal{Z}$.

B. Motivation of the TDF for HSI-MSI Fusion

The primary motivation which underpins the proposed CTDF method is that the TDF decomposition has superior feature extraction capability over the TR decomposition as well as 2-D spatial features extraction capability.

1) Feature Extraction Capability: Here, the proposed TDF decomposition is compared with the TR decomposition in terms of feature extraction capability. We evaluate the TDF-ALS algorithm and compare it with TR-ALS [67] using the Columbia Object Image Libraries (COIL)-100 dataset.[1] This dataset comprises 7200 color images of 100 objects, with 72 images per object, exhibiting various reflectance properties and complex geometric characteristics. Each image is a third-order tensor with dimensions $128 \times 128 \times 3$. We downsample each image to $32 \times 32 \times 3$. The dataset is then structured as a third-order tensor $\mathcal{Z}$ with dimensions $32^2 \times 3 \times 7200$. When applying TR-ALS with predefined TR-rank $[r_1, r_2, r_3]$ to tensor $\mathcal{Z}$, it generates three third-order TR cores, $\mathcal{G}_i$ for $i = 1, 2, 3$. Among these cores, the third

[1] https://www.kaggle.com/jessicali9530/coil100

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:24:23 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF THE TR-ALS AND TDF-ALS FOR CLASSIFICATION WITH DIFFERENT TDF-RANK AND TR-RANK (BOLD: BEST)

Algorithm	$m=2$	$m=3$
	Accuray (%)	Accuray (%)
TR-ALS	90.29	98.29
TDF-ALS	**93.07**	**99.47**

TR core, $\mathcal{G}_3 \in \mathbb{R}^{r_3\times 7200\times r_1}$, is used as the latent TR features, while the subchain $\mathcal{G}^{\neq 3}$ serves as the basis of the latent subspace. In the case of the proposed TDF-ALS algorithm, the third-order factor $\mathcal{D} \in \mathbb{R}^{r_2\times 7200\times r_1}$ is employed as the latent TDF features, and the fourth-order factor $\mathcal{C} \in \mathbb{R}^{r_1\times 32^2\times 3\times r_2}$ is considered as the latent subspace. In this experiment, for both TR-ALS and TDF-ALS, we set $r_1 = r_2 = r_3 = m$, and the range of m varies from 2 to 3. The relative error[2] and the maximum number of iterations are set to 10^{-5} and 10, respectively. We apply the TR-ALS and the TDF-ALS to the tensor $\mathcal{Z}$. The resulting TR core $\mathcal{G}_3$ and TDF factor $\mathcal{D}$ are utilized as the extracted features. By permutation and reshaping, we generate the corresponding feature matrices as $\mathbf{G}_3 \in \mathbb{R}^{7200\times m^2}$ and $\mathbf{D} \in \mathbb{R}^{7200\times m^2}$, respectively. Subsequently, we employ the k-nearest neighbor algorithm[3] with $k = 1$ for classification. The results are presented in Table I, which shows that the proposed TDF-ALS outperforms the TR-ALS. This underscores the superior feature extraction capability of TDF compared to TR decomposition. The rationale behind the superior capability of the proposed TDF to capture more complex features over the TR decomposition is due to the fourth-order factor of the TDF, which is higher than $\mathcal{Z}$.

2) Two-Dimensional Spatial Features Extraction Capability: As depicted in Fig. 1(a)–(c), the TDs employed by the existing CTD-based methods tend to separate 2-D spatial features in HR-HSI into two components, specifically the height features and the width features. Therefore, these CTD-based methods inevitably result in a partial loss of spatial features in the HR-HSI, which adversely affects fusion accuracy. The proposed TDF decomposition can address this issue. This format can extract 2-D spatial features using the fourth-order spatial factor $\mathcal{C} \in \mathbb{R}^{r_1\times H\times W\times r_2}$, as illustrated in Fig. 1(d).

C. Proposed CTDF Method for HSI-MSI Fusion

Based on the TDF decomposition (8), the third-order HR-HSI $\mathcal{X} \in \mathbb{R}^{H\times W\times S}$ can be represented as

$$\mathcal{X} = \mathcal{C} \times_{4,1}^{1,3} \mathcal{D} \tag{15}$$

where the tensor $\mathcal{C} \in \mathbb{R}^{r_1\times H\times W\times r_2}$ and $\mathcal{D} \in \mathbb{R}^{r_2\times S\times r_1}$ denote the spatial and spectral factors, respectively. In this expression, the spatial features are incorporated into $\mathcal{C}$, the spectral features are incorporated into $\mathcal{D}$, and the features of the three modes are incorporated into the unified model (15).

The HSI $\mathcal{Y}$ is a spatially degraded version of the HR-HSI $\mathcal{X}$. Assuming that the point spread function (PSF) of the hyperspectral sensor and the degrading matrices of the height mode and width mode is separable, then we may write

$$\mathcal{Y} = \mathcal{X} \times_2 \mathbf{P}_1 \times_3 \mathbf{P}_2 \tag{16}$$

where $\mathbf{P}_1 \in \mathbb{R}^{h\times H}$ and $\mathbf{P}_2 \in \mathbb{R}^{w\times W}$ are the degrading matrices along the height mode and the width mode, respectively. This describes the spatial response of the imaging sensors. Under the separability assumption, substituting (15) into (16), the HSI can be written as

$$\begin{aligned}\mathcal{Y} &= (\mathcal{C} \times_2 \mathbf{P}_1 \times_3 \mathbf{P}_2) \times_{4,1}^{1,3} \mathcal{D} \\ &= \widehat{\mathcal{C}} \times_{4,1}^{1,3} \mathcal{D}\end{aligned} \tag{17}$$

where $\widehat{\mathcal{C}} = \mathcal{C} \times {}_2\mathbf{P}_1 \times_3 \mathbf{P}_2 \in \mathbb{R}^{r_1\times h\times w\times r_2}$ is the degraded spatial factor.

Remark 2: The degraded spatial factor $\widehat{\mathcal{C}} = \mathcal{C} \times {}_2\mathbf{P}_1 \times_3 \mathbf{P}_2$ satisfies the following equation:

$$\widehat{\mathbf{C}}_{[\mathbf{m}_1;2]}^T = (\mathbf{P}_2 \otimes \mathbf{P}_1)\mathbf{C}_{[\mathbf{m}_1;2]}^T \tag{18}$$

where $\mathbf{m}_1 = [4, 1, 2, 3]$ is consistent in (9).

The MSI $\mathcal{Z}$ is the spectrally degraded version of $\mathcal{X}$, i.e.,

$$\mathcal{Z} = \mathcal{X} \times_3 \mathbf{P}_3 \tag{19}$$

where $\mathbf{P}_3 \in \mathbb{R}^{s\times S}$ is the degrading matrix of the spectrum mode. Substituting (15) into (19), MSI can also be represented as

$$\begin{aligned}\mathcal{Z} &= \mathcal{C} \times_{4,1}^{1,3} (\mathcal{D} \times_2 \mathbf{P}_3) \\ &= \mathcal{C} \times_{4,1}^{1,3} \widehat{\mathcal{D}}\end{aligned} \tag{20}$$

where $\widehat{\mathcal{D}} = \mathcal{D} \times {}_2\mathbf{P}_3 \in \mathbb{R}^{r_2\times s\times r_1}$ is the degraded spectral factor.

Remark 3: The degraded spectral factor $\widehat{\mathcal{D}} = \mathcal{D} \times {}_2\mathbf{P}_3$ satisfies the following equation:

$$\widehat{\mathbf{D}}_{[\mathbf{m}_2;2]} = \mathbf{D}_{[\mathbf{m}_2;2]}\mathbf{P}_3^T \tag{21}$$

where $\mathbf{m}_2 = [1, 3, 2]$ is consistent with (9).

The TDF degradation model from the HR-HSI to both MSI and HSI is illustrated in Fig. 2. Obviously, to generate the HR-HSI, we need to estimate the spatial factor $\mathcal{C}$ and spectral factor $\mathcal{D}$.

Taking (17) and (20) into account, we formulate the HSI-MSI fusion problem as the following least-squared optimization problem:

$$\min_{\mathcal{C},\mathcal{D}} \frac{1}{2}\left\|\mathcal{Y} - \widehat{\mathcal{C}} \times_{4,1}^{1,3} \mathcal{D}\right\|_F^2 + \frac{1}{2}\left\|\mathcal{Z} - \mathcal{C} \times_{4,1}^{1,3} \widehat{\mathcal{D}}\right\|_F^2 \tag{22}$$

where $\|\cdot\|_F$ denotes the Frobenius norm. We refer to model (22) as the CTDF model, denoted as $f(\mathcal{C}, \mathcal{D})$. This model is nonconvex, and the solutions for $\mathcal{C}$ and $\mathcal{D}$ are not unique. However, the objective function $f(\mathcal{C}, \mathcal{D})$ is convex for each block of variables when keeping other variables fixed. Therefore, we employ the PAM scheme [68], [69] to solve the CTDF model, which can be guaranteed to converge to a critical point. Specifically, $\mathcal{C}$ and $\mathcal{D}$ are iteratively updated as follows:

$$\begin{cases}\mathcal{C}^{k+1} = \arg\min\limits_{\mathcal{C}} f(\mathcal{C}, \mathcal{D}^k) + \dfrac{\rho}{2}\left\|\mathcal{C} - \mathcal{C}^k\right\|_F^2 \\ \mathcal{D}^{k+1} = \arg\min\limits_{\mathcal{D}} f(\mathcal{C}^{k+1}, \mathcal{D}) + \dfrac{\rho}{2}\left\|\mathcal{D} - \mathcal{D}^k\right\|_F^2\end{cases} \tag{23}$$

[2]Relative error $= \|\mathcal{X} - \mathcal{X}_1\|_F / \|\mathcal{X}\|_F$, where $\mathcal{X}$ and $\mathcal{X}_1$ denote the original tensor and reconstructed tensor, respectively.

[3]https://en.wikipedia.org/wiki/K-nearest_neighbors_algorithm

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:24:23 UTC from IEEE Xplore. Restrictions apply.

where ρ is a positive hyperparameter and $(\cdot)^k$ denotes the estimated blocks of variables in the kth iteration. Next, we present the optimizations for $\mathcal{C}$ and $\mathcal{D}$ in detail.

1) Optimization With Respect to $\mathcal{C}$: With $\mathcal{D}$ fixed, the optimization with respect to $\mathcal{C}$ in (23) can be written as

$$\arg\min_{\mathcal{C}} \frac{1}{2}\left\|\mathcal{Y}-\widehat{\mathcal{C}}\times_{4,1}^{1,3}\mathcal{D}^k\right\|_F^2 + \frac{1}{2}\left\|\mathcal{Z}-\mathcal{C}\times_{4,1}^{1,3}\widehat{\mathcal{D}}^k\right\|_F^2+\frac{\rho}{2}\left\|\mathcal{C}-\mathcal{C}^k\right\|_F^2 \quad (24)$$

where $\widehat{\mathcal{D}}^k=\mathcal{D}^k\times_2\mathbf{P}_3$. Taking Theorem 1, Remarks 2 and 3 into account, (24) can be written as

$$\begin{aligned}\arg\min_{\mathbf{C}_{[\mathbf{m}_1;2]}^T}\ &\frac{1}{2}\left\|\mathbf{Y}_{\langle 2\rangle}-\widehat{\mathbf{C}}_{[\mathbf{m}_1;2]}^T\mathbf{D}_{[\mathbf{m}_2;2]}^k\right\|_F^2\\&+\frac{1}{2}\left\|\mathbf{Z}_{\langle 2\rangle}-\mathbf{C}_{[\mathbf{m}_1;2]}^T\widehat{\mathbf{D}}_{[\mathbf{m}_2;2]}^k\right\|_F^2\\&+\frac{\rho}{2}\left\|\mathbf{C}_{[\mathbf{m}_1;2]}^T-\left(\mathbf{C}_{[\mathbf{m}_1;2]}^k\right)^T\right\|_F^2\end{aligned} \quad (25)$$

where $\widehat{\mathbf{C}}_{[\mathbf{m}_1;2]}^T=(\mathbf{P}_2\otimes\mathbf{P}_1)\mathbf{C}_{[\mathbf{m}_1;2]}^T$, $\widehat{\mathbf{D}}_{[\mathbf{m}_2;2]}^k=\mathbf{D}_{[\mathbf{m}_2;2]}^k\mathbf{P}_3^T$, $\mathbf{m}_1=[4,1,2,3]$, and $\mathbf{m}_2=[1,3,2]$.

Problem (25) is quadratic and its unique solution amounts to compute the following problem:

$$\mathbf{H}_1\mathbf{C}_{[\mathbf{m}_1;2]}^T\mathbf{H}_2+\mathbf{C}_{[\mathbf{m}_1;2]}^T\mathbf{H}_3=\mathbf{H}_4 \quad (26)$$

where

$$\begin{cases}\mathbf{H}_1=(\mathbf{P}_2\otimes\mathbf{P}_1)^T(\mathbf{P}_2\otimes\mathbf{P}_1)\\ \mathbf{H}_2=\mathbf{D}_{[\mathbf{m}_2;2]}^k\left(\mathbf{D}_{[\mathbf{m}_2;2]}^k\right)^T\\ \mathbf{H}_3=\mathbf{D}_{[\mathbf{m}_2;2]}^k\mathbf{P}_3^T\left(\mathbf{D}_{[\mathbf{m}_2;2]}^k\mathbf{P}_3^T\right)^T+\rho\mathbf{I}\end{cases}$$

and $\mathbf{H}_4=(\mathbf{P}_2\otimes\mathbf{P}_1)^T\mathbf{Y}_{\langle 2\rangle}(\mathbf{D}_{[\mathbf{m}_2;2]}^k)^T+\mathbf{Z}_{\langle 2\rangle}(\mathbf{D}_{[\mathbf{m}_2;2]}^k\mathbf{P}_3^T)^T+\rho(\mathbf{C}_{[\mathbf{m}_1;2]}^k)^T$. We use the conjugate gradient (CG) method [70] to solve (26). Then, $\mathbf{C}_{[\mathbf{m}_1;2]}=(\mathbf{C}_{[\mathbf{m}_1;2]}^T)^T$ and $\mathcal{C}=\text{Fold}_{[\mathbf{m}_1;2]}(\mathbf{C}_{[\mathbf{m}_1;2]})$.

2) Optimization With Respect to $\mathcal{D}$: With $\mathcal{C}$ fixed, the optimization with respect to $\mathcal{D}$ in (23) can be written as

$$\arg\min_{\mathcal{D}} \frac{1}{2}\left\|\mathcal{Y}-\widehat{\mathcal{C}}^{k+1}\times_{4,1}^{1,3}\mathcal{D}\right\|_F^2 + \frac{1}{2}\left\|\mathcal{Z}-\mathcal{C}^{k+1}\times_{4,1}^{1,3}\widehat{\mathcal{D}}\right\|_F^2+\frac{\rho}{2}\left\|\mathcal{D}-\mathcal{D}^k\right\|_F^2 \quad (27)$$

where $\widehat{\mathcal{C}}^{k+1}=\mathcal{C}^{k+1}\times_1\mathbf{P}_1\times_2\mathbf{P}_2$. Taking Theorem 1 and Remarks 2 and 3 into account, (27) can be written as

$$\begin{aligned}\arg\min_{\mathbf{D}_{[\mathbf{m}_2;2]}}\ &\frac{1}{2}\left\|\mathbf{Y}_{\langle 2\rangle}-\left(\widehat{\mathbf{C}}_{[\mathbf{m}_1;2]}^{k+1}\right)^T\mathbf{D}_{[\mathbf{m}_2;2]}\right\|_F^2\\&+\frac{1}{2}\left\|\mathbf{Z}_{\langle 2\rangle}-\left(\mathbf{C}_{[\mathbf{m}_1;2]}^{k+1}\right)^T\widehat{\mathbf{D}}_{[\mathbf{m}_2;2]}\right\|_F^2\\&+\frac{\rho}{2}\left\|\mathbf{D}_{[\mathbf{m}_2;2]}-\mathbf{D}_{[\mathbf{m}_2;2]}^k\right\|_F^2\end{aligned} \quad (28)$$

where $(\widehat{\mathbf{C}}_{[\mathbf{m}_1;2]}^{k+1})^T=(\mathbf{P}_2\otimes\mathbf{P}_1)(\mathbf{C}_{[\mathbf{m}_1;2]}^{k+1})^T$, $\widehat{\mathbf{D}}_{[\mathbf{m}_2;2]}=\mathbf{D}_{[\mathbf{m}_2;2]}\mathbf{P}_3^T$, $\mathbf{m}_1=[4,1,2,3]$, and $\mathbf{m}_2=[1,3,2]$.

Similar to (25), (28) is also quadratic and its unique solution amounts to compute the following problem:

$$\mathbf{M}_1\mathbf{D}_{[\mathbf{m}_2;2]}\mathbf{M}_2+\mathbf{M}_3\mathbf{D}_{[\mathbf{m}_2;2]}=\mathbf{M}_4 \quad (29)$$

Algorithm 2 CTDF-Based HSI-MSI Fusion

Input: HSI $\mathcal{Y}$, MSI $\mathcal{Z}$, the predefined TDF-rank $\mathbf{r}$, the positive hyperparameter ρ, the number of iterations K.
Output: HR-HSI $\mathcal{X}$.

```
Initialize k = 0, X^0 = imresize(Y, [H, W, S], 'bicubic'), C^0 and D^0 by Algorithm 1 using X^0.
while k < K do
    Update (C^{k+1}_{[m1;2]})^T via (26) with CG.
    C^{k+1}_{[m1;2]} = ((C^{k+1}_{[m1;2]})^T)^T.
    C^{k+1} = Fold_{[m1;2]}(C^{k+1}_{[m1;2]}).
    Update D^{k+1}_{[m2;2]} via (29) with CG.
    D^{k+1} = Fold_{[m2;2]}(D^{k+1}_{[m2;2]}).
    Update k = k + 1.
end while
Obtain HR-HSI X via (15).
```

where

$$\begin{cases}\mathbf{M}_1=\mathbf{C}_{[\mathbf{m}_1;2]}^{k+1}\left(\mathbf{C}_{[\mathbf{m}_1;2]}^{k+1}\right)^T\\ \mathbf{M}_2=\mathbf{P}_3^T\mathbf{P}_3\\ \mathbf{M}_3=\left((\mathbf{P}_2\otimes\mathbf{P}_1)\left(\mathbf{C}_{[\mathbf{m}_1;2]}^{k+1}\right)^T\right)^T(\mathbf{P}_2\otimes\mathbf{P}_1)\left(\mathbf{C}_{[\mathbf{m}_1;2]}^{k+1}\right)^T+\rho\mathbf{I}\end{cases}$$

and $\mathbf{M}_4=((\mathbf{P}_2\otimes\mathbf{P}_1)(\mathbf{C}_{[\mathbf{m}_1;2]}^{k+1})^T)^T\mathbf{Y}_{\langle 2\rangle}+\mathbf{C}_{[\mathbf{m}_1;2]}^{k+1}\mathbf{Z}_{\langle 2\rangle}\mathbf{P}_3+\rho\mathbf{D}_{[\mathbf{m}_2;2]}^k$. Similar to (26), we use the CG method to solve (29). Then, $\mathcal{D}^{k+1}=\text{Fold}_{[\mathbf{m}_2;2]}(\mathbf{D}_{[\mathbf{m}_2;2]}^{k+1})$.

We summarize the proposed CTDF method for HSI-MSI fusion in Algorithm 2, which aims at solving the CTDF model (22) by the PAM scheme shown in (23). Therein, we adopt the TDF-ALS (see Algorithm 1) to initialize the spatial factor $\mathcal{C}$ and the spectral factor $\mathcal{D}$, where we apply the cubic interpolation to the HSI to generate the input third-order tensor. The flowchart of the proposed CTDF method for HSI-MSI fusion is shown in Fig. 3.

D. Computational Complexity Analysis of Algorithm 2

In the CTDF, we optimize the spatial factor $\mathcal{C}$ and the spectral factor $\mathcal{D}$ by the CG algorithm. In each iteration of CG, the heaviest step is the multiplication of the system matrix by a vector. For the optimization with respect to $\mathbf{C}_{[\mathbf{m}_1;2]}$, the time complexity is $\mathcal{O}((HW)^2(r_2r_1)^2)$, and with respect to $\mathbf{D}_{[\mathbf{m}_2;2]}$, the time complexity is $\mathcal{O}((r_2r_1)^2S^2)$. These operations may be carried out efficiently in a matrix representation, reducing the complexity to $\mathcal{O}((HW)^2r_1r_2+HW(r_1r_2)^2)$ and $\mathcal{O}((r_1r_2)^2S+(r_1r_2)S^2)$, for the optimizations of $\mathbf{C}_{[\mathbf{m}_1;2]}$ and $\mathbf{D}_{[\mathbf{m}_2;2]}$, respectively. Thus, the total time complexity of each CTDF iteration is

$$\begin{aligned}\mathcal{O}\big(&N_{\text{CG1}}\big((HW)^2r_1r_2\\&+HW(r_1r_2)^2\big)+N_{\text{CG2}}\big((r_1r_2)^2S+(r_1r_2)S^2\big)\big)\end{aligned} \quad (30)$$

where N_{CG1} and N_{CG2} denote the number of CG iterations for updating $\mathbf{C}[\mathbf{m}_1;2]$ and $\mathbf{D}[\mathbf{m}_2;2]$, respectively. In our experiments, we found that 30 iterations yield a good approximation of the solution for (26) and (29). Therefore, we set $N_{\text{CG1}}=N_{\text{CG2}}=30$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:24:23 UTC from IEEE Xplore. Restrictions apply.

E. Convergence Analysis of Algorithm 2

Theorem 3: Consider the iterative process governed by Algorithm 2, generating a sequence $\{(\mathcal{C}^k, \mathcal{D}^k)\}_{k\in\mathbb{N}}$. Under the prescribed conditions of Algorithm 2, the sequence converges globally to a critical point $\{(\mathcal{C}^*, \mathcal{D}^*)\}$ of the objective function $f(\mathcal{C}, \mathcal{D})$ as delineated by (22).

The proof of Theorem 3 is detailed in the Appendix.

VI. Experiments

A. Datasets

1) *CAVE Dataset*[4]*:* The CAVE dataset consists of 31 indoor HR-HSIs, each comprising 31 spectral bands and possessing a spatial resolution of 512 × 512 pixels. From this dataset, we randomly selected one HR-HSI, called *paints*. We resized *paints* to 256 × 256 × 31 to serve as a ground truth (GT). We generated an HSI with a size of 64 × 64 × 31 by downsampling the GT through averaging of 4 × 4 disjoint spatial blocks. A Nikon D700 camera was used to generate an MSI with a size of 256 × 256 × 3.
2) *Harvard Dataset*[5]*:* The Harvard dataset comprises 50 HR-HSIs that captured both outdoor and indoor scenes illuminated by daylight. Each HR-HSI contains 31 spectral bands and has a spatial resolution of 1392 × 1040 pixels. We randomly selected one HR-HSI, i.e., *imgb0*, and extracted the up-left cube of size 512 × 512 × 31. We then resized this cube to 256 × 256 × 31 as a GT. The HSI of size 64 × 64 × 31 was generated by downsampling the GT through averaging of 4 × 4 disjoint spatial blocks. In addition, we captured an MSI with three spectral bands using a Nikon D700 camera.
3) *Pavia University (PU) Dataset*[6]*:* The PU dataset originally had a size of 610 × 340 × 115. We reduced the number of spectral bands to 93 by eliminating the bands associated with water vapor absorption and selected the up-left cube of size 256 × 256 × 93 as a GT. The HSI of size 64 × 64 × 93 was generated by downsampling the GT through averaging 4 × 4 disjoint spatial blocks. An IKONOS-like spectral reflectance response filter [51] was used to generate an MSI with a size of 256 × 256 × 4.
4) *Chikusei Dataset*[7] *[71]:* The Chikusei dataset was captured using the Headwall Hyperspec-VNIR-C imaging sensor. It comprises 128 bands within the spectral range from 363 to 1018 nm. The scene dimensions are 2517 × 2335 pixels, with a ground sampling distance (GSD) of 2.5 m. A subimage of size 256 × 256 × 128 is chosen as a GT. The HSI was generated by applying a 7 × 7 Gaussian filter (standard deviation 2) and subsequently subsampling the image by a factor of 4. The MSI was simulated by averaging every 32 spectral bands.

[4]https://www.cs.columbia.edu/CAVE/databases/multispectral/
[5]http://vision.seas.harvard.edu/hyperspec/d2 × 5g3/
[6]https://rslab.ut.ac.ir/data
[7]http://naotoyokoya.com/Download.html

B. Compared Methods and Quantitative Metrics

1) Compared Methods: In this study, we conducted a comparative analysis of the proposed CTDF method against seven state-of-the-art methods. These include UMF-based method S^4-LRR [53]; CMF-based method CNMF [54]; UTD-based methods IR-TenSR [59] and LTTR [60]; CTD-based methods CSTF [16] and CTRF [17]; and a CNN-based method ZSL [44]. The implementation codes for these methods were obtained from the authors' websites, and all model parameters underwent fine-tuning in accordance with the authors' recommended settings. All the methods were implemented in MATLAB R2020a on a PC with 12nd Gen Intel[8] Core[9] i9-12900K at 3.20 GHz and 64 GB RAM, except for the ZSL [44], which was implemented in PyTorch and run on NVIDIA 1080Ti GPU.

2) Quantitative Metrics: The quality of the fused HR-HSI was assessed using various quantitative metrics. Six metrics were employed in our study: peak signal-to-noise ratio (PSNR), structural similarity index (SSIM) [72], relative dimensionless global error in synthesis (ERGAS) [73], spectral angle mapper (SAM) [74], and root mean square error (RMSE) [16]. Higher PSNR and SSIM values indicate better quality, while lower ERGAS, SAM, and RMSE values are associated with higher quality. We also compared the computational efficiency of the methods by measuring the runtime in seconds. To ensure consistency, all image values were scaled to the range $[0, 255]$ when calculating the quantitative metrics.

C. Experimental Results on Simulated Data Fusion

1) Quantitative Comparison: The results on the CAVE, the Harvard, the PU, and the Chikusei datasets of the methods are presented in Table II. Because the ZSL method was run on GPU with a training phase and the other methods were executed on CPU without the training phase, the runtime of the ZSL method is not included in the results. As depicted in Table II, the proposed CTDF method achieves better fusion quality results than other compared methods. On the CAVE, the PU, and the Chikusei datasets, the CNMF method has the lowest runtime, and the CTDF achieves the lowest runtime on the Harvard dataset. The CNMF exhibits the lowest runtime because of its straightforward matrix update procedure. The UMF-based method S^4-LRR generates the highest runtime on the CAVE, the PU, and the Chikusei due to its laborious variable updating. Compared to TD-based methods and S^4-LRR, the proposed CTDF stands out with the shortest runtime. The results shown in Table II illustrate the effectiveness and efficiency of the proposed CTDF method.

2) Visual Comparison: To offer a comprehensive evaluation of the fused results among the methods, we extend the fusion quality assessment to visual perception. Pseudo-color images that are reconstructed from the spectral bands along with their corresponding error images from the CAVE, the Harvard, the PU, and the Chikusei datasets are shown in Figs. 4–7, respectively. The representative regions in these images are highlighted by red and green boxes. The CNN-based method,

[8]Registered trademark.
[9]Trademarked.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:24:23 UTC from IEEE Xplore. Restrictions apply.

TABLE II

RESULTS FOR SIX METRICS ACROSS ALL METHODS ARE PRESENTED FOR THE CAVE, HARVARD, PU, AND CHIKUSEI DATASETS. ALL METHODS RUN ON CPU, EXCEPT FOR THE CNN-BASED METHOD ZSL [44], WHICH RUNS ON GPU (BOLD: BEST; UNDERLINE: SECOND BEST)

Dataset	Method	PSNR ↑	SSIM ↑	ERGAS ↓	SAM ↓	RMSE ↓	Time (s) ↓
CAVE	CNMF [54]	37.853	0.985	1.857	4.138	3.600	**3.648**
	CSTF [16]	42.900	0.990	1.410	3.567	2.657	39.692
	LTTR [60]	33.998	0.977	2.728	3.631	5.328	126.039
	S^4-LRR [53]	41.523	0.993	1.264	2.691	2.450	146.156
	IR-TenSR [59]	33.913	0.976	2.750	3.895	5.371	12.621
	CTRF [17]	41.821	0.986	1.323	4.327	2.552	40.789
	ZSL [44]	38.082	0.986	1.955	4.930	3.751	–
	CTDF (Ours)	**45.525**	**0.995**	**0.997**	**2.259**	**1.944**	7.342
Harvard	CNMF [54]	48.725	0.995	0.741	1.245	1.232	49.363
	CSTF [16]	51.270	0.993	0.635	1.612	1.170	78.837
	LTTR [60]	44.653	0.988	1.476	1.575	2.051	113.645
	S^4-LRR [53]	51.689	0.995	0.953	1.300	0.946	110.354
	IR-TenSR [59]	44.438	0.988	1.765	1.547	2.034	11.750
	CTRF [17]	50.794	0.994	0.647	1.448	1.092	35.685
	ZSL [44]	51.801	0.996	0.604	1.460	0.976	–
	CTDF (Ours)	**54.323**	**0.997**	**0.455**	**1.034**	**0.746**	**11.275**
PU	CNMF [54]	41.833	0.991	1.232	1.785	2.171	**18.079**
	CSTF [16]	43.936	0.989	0.969	1.780	1.771	47.003
	LTTR [60]	37.893	0.954	2.550	4.327	4.823	98.850
	S^4-LRR [53]	38.250	0.970	1.878	3.412	3.620	273.455
	IR-TenSR [59]	37.555	0.969	2.146	3.675	3.976	52.155
	CTRF [17]	43.041	0.987	1.147	2.160	2.101	38.570
	ZSL [44]	42.615	0.989	1.205	2.158	2.127	–
	CTDF (Ours)	**45.468**	**0.993**	**0.856**	**1.483**	**1.589**	29.422
Chikusei	CNMF [54]	49.483	0.996	**2.414**	1.251	0.931	**6.769**
	CSTF [16]	50.693	0.995	2.549	1.265	0.830	75.003
	LTTR [60]	41.472	0.981	4.557	2.617	2.415	97.537
	S^4-LRR [53]	48.369	0.993	2.620	1.656	1.150	168.436
	IR-TenSR [59]	41.037	0.978	4.966	2.806	2.498	30.700
	CTRF [17]	50.062	0.995	2.727	1.319	0.889	64.973
	ZSL [44]	50.356	**0.997**	2.771	**1.075**	0.889	–
	CTDF (Ours)	**50.927**	0.996	2.622	1.108	**0.810**	19.113

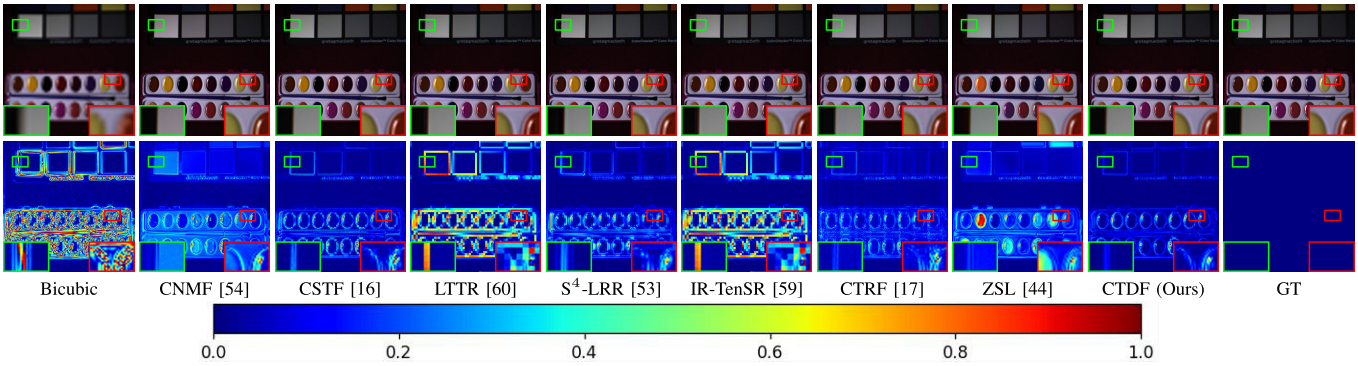


Fig. 4. (First row) Pseudo-color images consisting of the 30th, 20th, and 10th bands of the fused *paints*, generated by bicubic upsampling, CNMF [54], CSTF [16], LTTR [60], S^4-LRR [53], IR-TenSR [59], CTRF [17], ZSL [44], CTDF, and GT. (Second row) Error images corresponding to the first row, enhanced by multiplying by a fixed number (i.e., 15).

ZSL, demonstrates better visual performance compared to the MF-based methods (CNMF and S^4-LRR) as well as UTD-based methods (LTTR and IR-TenSR). However, it exhibits suboptimal performance compared to the CTD-based methods (CSTF, CTRF, and CTDF). It can be observed that the coupled methods, namely, CNMF, CSTF, CTRF, and the proposed CTDF, generally exhibit lower errors compared to the uncoupled decomposition-based methods such as LTTR, S^4-LRR, and IR-TenSR. The CTD-based methods, including CSTF, CTRF, and CTDF, consistently demonstrate quality image details compared to the CMF-based method (i.e., CNMF). However, among the CTD-based methods, the proposed CTDF achieves the least error, which demonstrates the advantages of the CTDF on the HSI-MSI fusion problem.

D. Experimental Results on Real-Data Fusion

For real-data experiments, the HSI was captured by the Hyperion imager [75] over the Cuprite mining district, Nevada, USA, on September 19, 2011. It consists of 167 bands spanning the VNIR and SWIR at a GSD of 30 m, and the scene dimensions are 927 × 379 pixels. The MSI for the SWIR was acquired by the WorldView-3 (WV3) satellite over

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:24:23 UTC from IEEE Xplore. Restrictions apply.

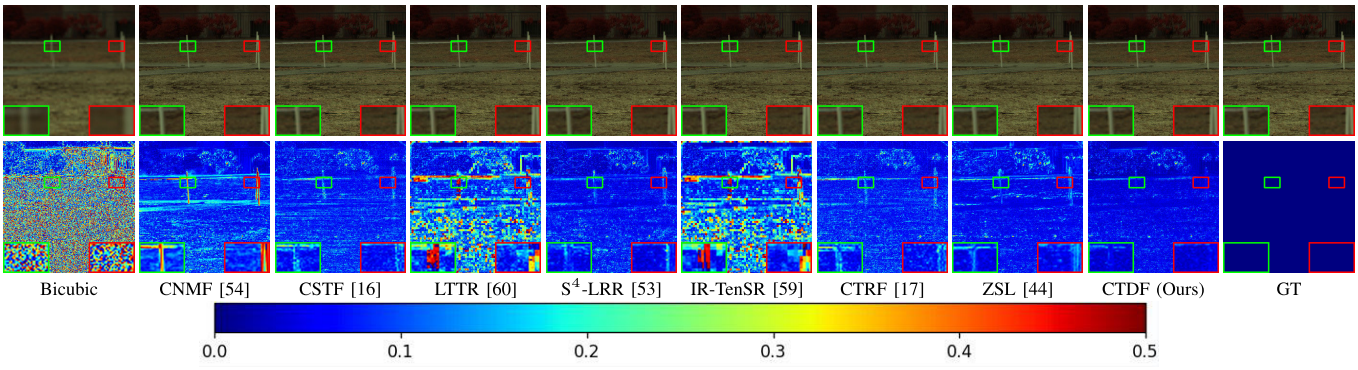


Fig. 5. (First row) Pseudo-color images consisting of the 30st, 20th, and 10st bands of the fused *imgb0*, generated by bicubic upsampling, CNMF [54], CSTF [16], LTTR [60], S^4-LRR [53], IR-TenSR [59], CTRF [17], ZSL [44], CTDF, and GT. (Second row) Error images corresponding to the first row, enhanced by multiplying by a fixed number (i.e., 20).

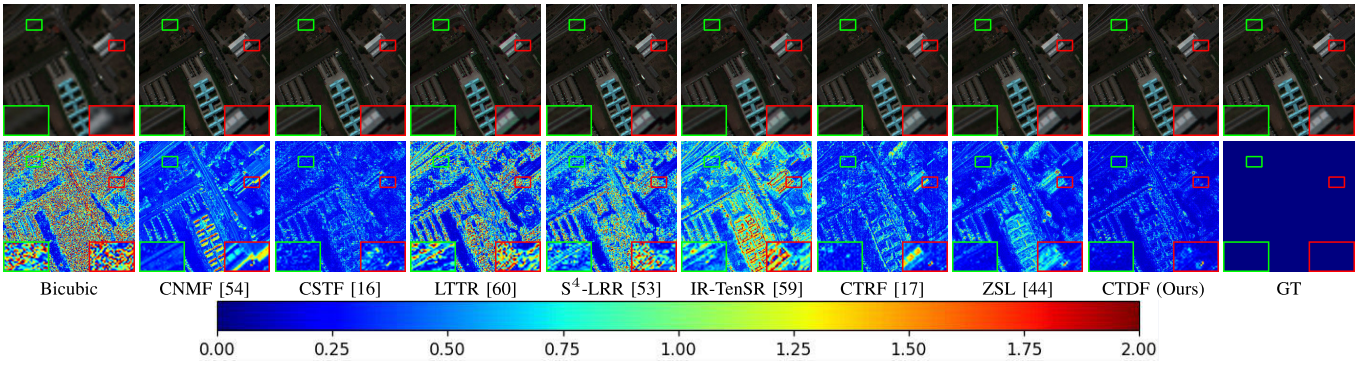


Fig. 6. (First row) Pseudo-color images consisting of the 40th, 20th, and 10th bands of the fused PU data, generated by bicubic upsampling, CNMF [54], CSTF [16], LTTR [60], S^4-LRR [53], IR-TenSR [59], CTRF [17], ZSL [44], CTDF, and GT. (Second row) Error images corresponding to the first row, enhanced by multiplying by a fixed number (i.e., 30).

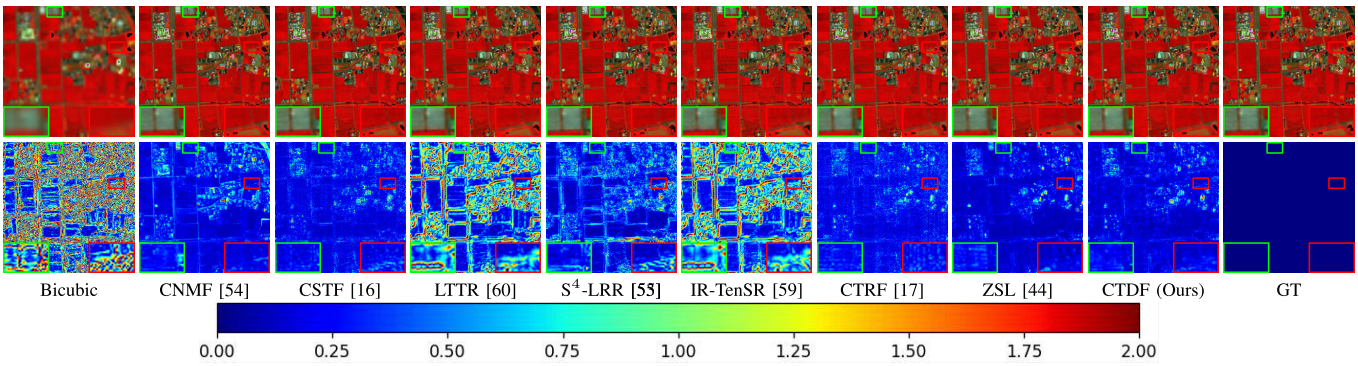


Fig. 7. (First row) Pseudo-color images consisting of the 120th, 60th, and 10th bands of the fused Chikusei data, generated by bicubic upsampling, CNMF [54], CSTF [16], LTTR [60], S^4-LRR [53], IR-TenSR [59], CTRF [17], ZSL [44], CTDF, and GT. (Second row) Error images corresponding to the first row, enhanced by multiplying by a fixed number (i.e., 20).

the same region resampled at 7.5 m GSD on September 19, 2014. The size of the MSI is 3710 × 1518 × 8. A subimage in the original HSI sized 128 × 128 × 167, and the corresponding registered subimage in the original MSI, sized 512 × 512 × 8, are selected as an HSI and an MSI, respectively. We estimated the spectral and spatial responses by the method proposed by Dian et al. [44]. Pseudo-color images obtained by fusing the Hyperion and the WV3 data using all the test methods are displayed in Fig. 8. The representative regions in these images are highlighted by red and green boxes. We also presented the runtime of each method in Fig. 8, excluding the ZSL method, as analyzed in Section VI-C1. From this figure, it is obvious that the proposed CTDF method, the ZSL, and the CNMF can provide high-quality image details and outperform the other compared approaches. Moreover, among the TD-based methods, the CTDF method exhibits the lowest runtime. The results are shown in Fig. 8 illustrate the effectiveness and efficiency of the proposed CTDF in real-data fusion.

VII. Discussion

A. Hyperparameter Analysis

The analysis of the impact of various hyperparameters of the Algorithm 2 is presented here. The hyperparameters under

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:24:23 UTC from IEEE Xplore. Restrictions apply.

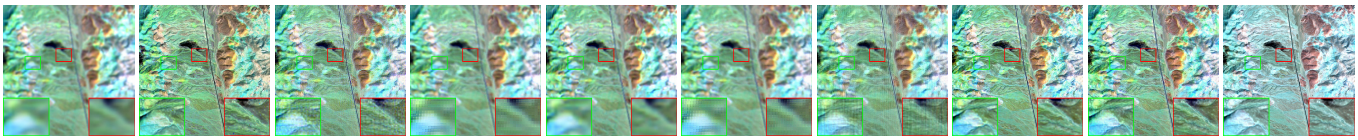


Fig. 8. Pseudo-color images consisting of the 2.20 μm red (152nd), 1.52 μm (103rd), and 0.57 μm (15th) bands of fused Hyperion and WV3 data, generated by Hyperion (HSI), CNMF [54], CSTF [16], LTTR [60], S^4-LRR [53], IR-TenSR [59], CTRF [17], ZSL [44], and CTDF, along with the runtime of all methods, excluding ZSL. WV3 (MSI) (the 2.32 μm (8th), 1.57 μm (2nd), and 1.21 μm (1st) bands) (bold: best; underline: second best).

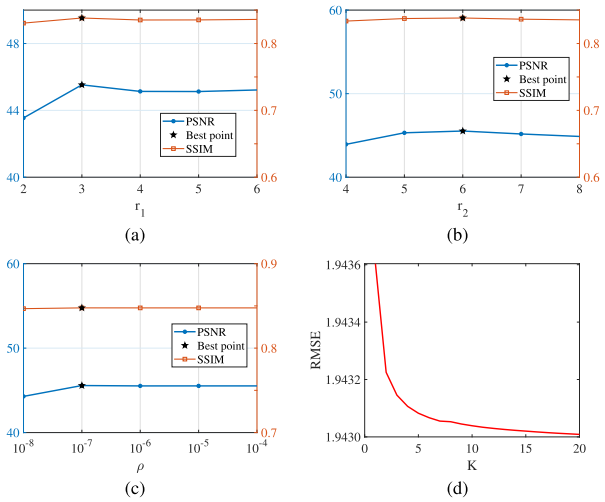


Fig. 9. Parameter analysis of Algorithm 2: (a) r_1, (b) r_2, (c) ρ, and (d) K (dataset: *paints* in CAVE dataset).

investigation include the TDF-rank $\mathbf{r} = [r_1, r_2]$, the ρ, and the number of iterations K. The CAVE dataset (refer to Section VI-A) is used in this discussion.

1) Analysis of the TDF-Rank $\mathbf{r}$*:* The tensor rank is primarily related to the intrinsic structure within the data. Determining the optimal tensor rank is a challenging task in the TN community. The tensor rank is generally treated as a hyperparameter in TN methods, and determining the optimal tensor rank often involves adjusting it based on multiple criteria, including but not limited to model accuracy. In this article, we employed a methodology based on PSNR and SSIM curves to generate the optimal TDF rank. The goal was to identify the optimal values for both r_1 and r_2. To explore these values, we considered candidate sets for r_1 and r_2 as $[2, 6]$ and $[4, 8]$, respectively. As shown in Fig. 9(a), we observed a consistent trend in the PSNR and the SSIM curves, characterized by a sharp increase within the range of 2–3, followed by a decrease and stability within 3–6. Thus, $r_1 = 3$ is selected as the optimal value for the CAVE dataset. Similarly, as shown in Fig. 9(b), the PSNR and the SSIM curves exhibited a shared pattern involving a significant increase within the range of 4–6, followed by a decrease within 6–8. Hence, $r_2 = 6$ is selected as the optimal choice for the CAVE dataset.

2) Analysis of the Hyperparameter ρ*:* To determine the optimal value for the ρ, we conducted experiments using a set of candidate values, specifically $[1e-8, 1e-4]$. By analyzing the PSNR and the SSIM curves, we observed that the behavior of both curves of the ρ varies, as depicted in Fig. 9(c). The PSNR curve experienced a sharp increase within the range of $1e-8$ to $1e-7$, then remained stable. Whereas the SSIM curve did not show obvious changes. Therefore, we selected $\rho = 1e-7$ for the CAVE dataset.

3) Analysis of the Number of Iterations K*:* Fig. 9(d) shows the variation curve of the RMSE as the number of iterations K increases. Our observations revealed that the proposed CTDF algorithm yielded satisfactory RMSE values at the first iteration, and the change in RMSE value from the third iteration to the 20th iteration remained within $2 \times 1e-4$. This demonstrates the high efficiency of the proposed CTDF method. Thus, we established $K = 3$ as the suitable choice for the CAVE dataset.

For the Harvard, PU, and Chikusei datasets, we employed the same strategy as used for the CAVE dataset to determine the optimal hyperparameters. In the real-data experiments, we determined the optimal hyperparameters based on visual performance. The suggested ranges for hyperparameters and the specific values selected for all five datasets are presented in Table III.

TABLE III
PARAMETER CHOICE OF THE PROPOSED METHOD ON DIFFERENT DATASETS

	r_1	r_2	ρ	K
Suggested Range	$[2, 9]$	$[4, 8]$	$[1e-8, 1e-4]$	$[3, 20]$
CAVE	3	6	10^{-7}	3
Harvard	7	4	10^{-7}	3
PU	8	8	10^{-5}	3
Chikusei	9	5	10^{-8}	3
Hperion-WV3	6	6	10^{-6}	3

B. Analysis of Downsampling Factor

In our investigations, both in simulated and real-data fusion experiments, we initially focused on a downsampling factor of 4. However, in this section, we broaden our perspective by examining the performance of the proposed CTDF method under different downsampling factors, specifically 4, 8, and 16. Fig. 10 illustrates the PSNR curves of the proposed CTDF method and other compared methods on the PU dataset with the aforementioned downsampling factors. Notably, as the downsampling factor increases, all PSNR curves exhibit a decreasing trend. This decline is particularly prominent in the case of the ZSL method, as its performance is adversely affected by the reduction in training data associated with higher downsampling factors. Despite these challenges, our proposed CTDF method consistently outperforms the comparative methods across the various downsampling factors.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:24:23 UTC from IEEE Xplore. Restrictions apply.

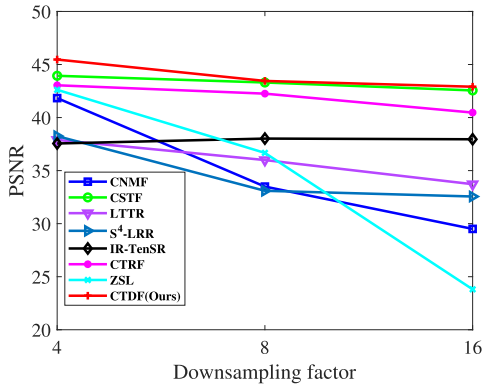


Fig. 10. PSNR curves of all eight methods on the PU dataset with downsampling factors of 4, 8, and 16.

This resilience underscores the effectiveness and robustness of the CTDF approach, even when faced with increased downsampling factors.

C. Analysis of the TDF Decomposition

As outlined in Section IV, we represented the third-order HR-HSI as a fourth-order spatial factor and a third-order spectral factor. Here, we discuss the rationale behind the representation of the third-order HR-HSI as the fourth-order spatial factor and the third-order spectral factor. Note that the dimension of the spectral factor depends on the dimension of the spatial factor, that is, $D(\text{spectral factor}) = \text{D(spatial factor)-1}$, where $D(\cdot)$ means the dimension operator. To preserve the 2-D spatial features of the HR-HSI, the spatial factor must be a tensor with a dimension higher than 2. Thus, we opted for third-, fourth-, and fifth-order spatial factors. Specifically, given a third-order tensor $\mathcal{X} \in \mathbb{R}^{H\times W\times S}$, we constructed the following three TDF formats.

1) $\mathcal{X} = \mathcal{C} \times {}^{1}_{3}\mathcal{D}$, where the spatial factor $\mathcal{C} \in \mathbb{R}^{H\times W\times r}$ and spectral factor $\mathcal{D} \in \mathbb{R}^{r\times S}$ are third- and second-order tensors, respectively.
2) $\mathcal{X} = \mathcal{C} \times {}^{1,3}_{4,1}\mathcal{D}$, where the spatial factor $\mathcal{C} \in \mathbb{R}^{r_1\times H\times W\times r_2}$ and spectral factor $\mathcal{D} \in \mathbb{R}^{r_2\times S\times r_1}$ are fourth- and third-order tensors, respectively.
3) $\mathcal{X} = \mathcal{C} \times {}^{1,2,4}_{4,3,1}\mathcal{D}$, where the spatial factor $\mathcal{C} \in \mathbb{R}^{r_1\times H\times r_2\times W\times r_3}$ and spectral factor $\mathcal{D} \in \mathbb{R}^{r_3\times r_2\times S\times r_1}$ are fifth- and fourth-order tensors, respectively.

We summarized these three TDF formats in Table IV. To assess the performance of the three TDFs, we employed them for HSI-MSI fusion. We formulated their corresponding CTDF models similar to the model (22) and designed PAM-based algorithms similar to the Algorithm 2 to solve the three models. The *paints* sample in the CAVE dataset is used as test data, and the six metrics presented in Section VI-B2 are used to evaluate their performance. The evaluation comparison is presented in Table V. In examining this table, it is obvious that (a) exhibits the lowest fusion accuracy. A plausible reason for this outcome is the absence of a higher order factor than that of HR-HSI. On the other hand, (c) has a better performance than (b) in terms of PSNR, SAM, and RMSE. However, the runtime of (c) is more than 6 times that of (b). Considering the fusion accuracy and computational efficiency, we selected (b) as the best TDF format.

TABLE IV
THREE TDF FORMATS OF A GIVEN THIRD-ORDER TENSOR $\mathcal{X} \in \mathbb{R}^{H\times W\times S}$

	Spatial Factor	Spectral Factor	TDF
(a)	$\mathcal{C} \in \mathbb{R}^{H\times W\times r}$	$\mathcal{D} \in \mathbb{R}^{r\times S}$	$\mathcal{X} = \mathcal{C} \times {}^{1}_{3}\mathcal{D}$
(b)	$\mathcal{C} \in \mathbb{R}^{r_1\times H\times W\times r_2}$	$\mathcal{D} \in \mathbb{R}^{r_2\times S\times r_1}$	$\mathcal{X} = \mathcal{C} \times {}^{1,3}_{4,1}\mathcal{D}$
(c)	$\mathcal{C} \in \mathbb{R}^{r_1\times H\times r_2\times W\times r_3}$	$\mathcal{D} \in \mathbb{R}^{r_3\times r_2\times S\times r_1}$	$\mathcal{X} = \mathcal{C} \times {}^{1,2,4}_{4,3,1}\mathcal{D}$

TABLE V
RESULTS OF SIX METRICS FOR THE THREE TDFS ON PAINTS IN THE CAVE DATASET (BOLD: BEST; UNDERLINE: SECOND BEST)

Method	PSNR ↑	SSIM ↑	ERGAS ↓	SAM ↓	RMSE ↓	Time (s) ↓
(a)	44.242	0.993	1.265	2.667	2.410	**6.380**
(b)	<u>45.525</u>	**0.995**	**0.997**	<u>2.259</u>	<u>1.944</u>	<u>7.342</u>
(c)	**46.052**	**0.995**	1.010	**2.246**	**1.943**	47.136

VIII. CONCLUSION

In this article, we introduced the CTDF method as a solution to the HSI-MSI fusion problem. We proposed the TDF decomposition, which represents the third-order tensor as a fourth-order spatial factor and a third-order spectral factor, connected through tensor contraction. Due to its higher order spatial factor than that of HR-HSI, the TDF decomposition exhibits better feature extraction capability than the TR decomposition. Moreover, it can extract 2-D spatial features using the fourth-order spatial factor. This capability is overlooked by other TDs. Based on these inherent advantages, we formulated the CTDF model. To solve this model, we designed a PAM-based algorithm and provided insights into its computational complexity and convergence analysis. We performed extensive experiments to select the optimal hyperparameters for the PAM-based algorithm. Furthermore, we analyzed the structure representation of the TDF decomposition. The experimental results demonstrate that the CTDF method attains a higher level of fusion accuracy compared to other state-of-the-art methods. Comparing the proposed CTDF method with other TD-based methods, the CTDF achieves the lowest runtime. This affirms the effectiveness and efficiency of the proposed CTDF method.

APPENDIX

Theorem 3: Consider the iterative process governed by Algorithm 2, generating a sequence $\{(\mathcal{C}^k, \mathcal{D}^k)\}_{k\in\mathbb{N}}$. Under the prescribed conditions of Algorithm 2, the sequence converges globally to a critical point $\{(\mathcal{C}^*, \mathcal{D}^*)\}$ of the objective function $f(\mathcal{C}, \mathcal{D})$ as delineated by (22).

Proof: According to the finite length property in [76], the sequence $\{(\mathcal{C}^k, \mathcal{D}^k)\}_{k\in\mathbb{N}}$ converges to the critical point $\{(\mathcal{C}^*, \mathcal{D}^*)\}$ of $f(\mathcal{C}, \mathcal{D})$, as a consequence of the satisfaction of the following four properties.

1) The objective function $f(\mathcal{C}, \mathcal{D})$ is a Kurdyka–Łojasiewicz (KŁ) function.
2) Considering the sequence $\{(\mathcal{C}^k, \mathcal{D}^k)\}_{k\in\mathbb{N}}$ generated by Algorithm 2, the sequence $\{f(\mathcal{C}^k, \mathcal{D}^k)\}_{k\in\mathbb{N}}$ explicitly fulfills the inequality

$$f(\mathcal{C}^k, \mathcal{D}^k) - f(\mathcal{C}^{k+1}, \mathcal{D}^{k+1}) \geq \frac{\rho}{2}\left(\|\mathcal{C}^{k+1} - \mathcal{C}^k\|_F^2 + \|\mathcal{D}^{k+1} - \mathcal{D}^k\|_F^2\right).$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:24:23 UTC from IEEE Xplore. Restrictions apply.

3) Let $\{(\mathcal{C}^k, \mathcal{D}^k)\}$ be the sequence generated by Algorithm 2. Then, there exists a constant $L_{\mathcal{C}}$ representing the Lipschitz constant for $\partial_{\mathcal{C}} f(\mathcal{C}, \mathcal{D})$, such that

$$\|\partial f(\mathcal{C}^{k+1}, \mathcal{D}^{k+1})\|_F \leq (L_{\mathcal{C}} + \rho)\big(\|\mathcal{C}^{k+1} - \mathcal{C}^k\|_F^2 + \|\mathcal{D}^{k+1} - \mathcal{D}^k\|_F^2\big).$$

4) The sequence $\{(\mathcal{C}^k, \mathcal{D}^k)\}_{k\in\mathbb{N}}$ generated by Algorithm 2 is bounded.

Proof of Property 1: According to [76] and [77], a semi-algebraic function is a KŁ function. $f(\mathcal{C}, \mathcal{D})$ consists of two Frobenius norm-based functions, namely, $\|\mathcal{Y} - \widehat{\mathcal{C}} \times_{4,1}^{1,3} \mathcal{D}\|_F^2$ and $\|\mathcal{Z} - \mathcal{C} \times_{4,1}^{1,3} \widehat{\mathcal{D}}\|_F^2$. These two functions are both real polynomial functions of two coupling variables, thus they are semi-algebraic. Consequently, $f(\mathcal{C}, \mathcal{D})$, being a finite sum of semi-algebraic functions, is a KŁ function.

Proof of Property 2: Since the sequences $\{\mathcal{C}^k\}_{k\in\mathbb{N}}$ and $\{\mathcal{D}^k\}_{k\in\mathbb{N}}$ are generated by (23), $\mathcal{C}^{k+1}$ and $\mathcal{D}^{k+1}$ are the minimizer of $(k+1)$th iteration of (23). Therefore, we have

$$\begin{cases} f(\mathcal{C}^{k+1}, \mathcal{D}^k) + \dfrac{\rho}{2}\big\|\mathcal{C}^{k+1} - \mathcal{C}^k\big\|_F^2 \leq f(\mathcal{C}^k, \mathcal{D}^k) \\ f(\mathcal{C}^{k+1}, \mathcal{D}^{k+1}) + \dfrac{\rho}{2}\big\|\mathcal{D}^{k+1} - \mathcal{D}^k\big\|_F^2 \leq f(\mathcal{C}^{k+1}, \mathcal{D}^k). \end{cases} \tag{31}$$

Then, the following inequality holds:

$$f(\mathcal{C}^k, \mathcal{D}^k) - f(\mathcal{C}^{k+1}, \mathcal{D}^{k+1}) \geq \frac{\rho}{2}\big(\|\mathcal{C}^{k+1} - \mathcal{C}^k\|_F^2 + \|\mathcal{D}^{k+1} - \mathcal{D}^k\|_F^2\big).$$

Proof of Property 3: Since $\mathcal{C}^{k+1}$, $\mathcal{D}^{k+1}$ are the optimal solution of $(k+1)$th iteration of (23). According to the first-order optimal conditions, we have

$$\begin{cases} 0 \in \partial_{\mathcal{C}} f(\mathcal{C}^{k+1}, \mathcal{D}^k) + \rho(\mathcal{C}^{k+1} - \mathcal{C}^k), \\ 0 \in \partial_{\mathcal{D}} f(\mathcal{C}^{k+1}, \mathcal{D}^{k+1}) + \rho(\mathcal{D}^{k+1} - \mathcal{D}^k). \end{cases} \tag{32}$$

Based on the subdifferentiability property, i.e.,

$$\partial f(\mathcal{C}^{k+1}, \mathcal{D}^{k+1}) = (\partial_{\mathcal{C}} f(\mathcal{C}^{k+1}, \mathcal{D}^{k+1}), \partial_{\mathcal{D}} f(\mathcal{C}^{k+1}, \mathcal{D}^{k+1})) \tag{33}$$

we can establish the following inequality:

$$\big\|\partial f(\mathcal{C}^{k+1}, \mathcal{D}^{k+1})\big\|_F \leq \big\|\partial_{\mathcal{C}} f(\mathcal{C}^{k+1}, \mathcal{D}^{k+1})\big\|_F + \big\|\partial_{\mathcal{D}} f(\mathcal{C}^{k+1}, \mathcal{D}^{k+1})\big\|_F. \tag{34}$$

Substitute (32) into (34) yields

$$\begin{aligned} &\big\|\partial f(\mathcal{C}^{k+1}, \mathcal{D}^{k+1})\big\|_F \\ &\leq \big\|\partial_{\mathcal{C}} f(\mathcal{C}^{k+1}, \mathcal{D}^{k+1}) - \partial_{\mathcal{C}} f(\mathcal{C}^{k+1}, \mathcal{D}^k) - \rho(\mathcal{C}^{k+1} - \mathcal{C}^k)\big\|_F \\ &+ \big\|\partial_{\mathcal{D}} f(\mathcal{C}^{k+1}, \mathcal{D}^{k+1}) - \partial_{\mathcal{D}} f(\mathcal{C}^{k+1}, \mathcal{D}^{k+1}) - \rho(\mathcal{D}^{k+1} - \mathcal{D}^k)\big\|_F \\ &\leq \big\|\partial_{\mathcal{C}} f(\mathcal{C}^{k+1}, \mathcal{D}^{k+1}) - \partial_{\mathcal{C}} f(\mathcal{C}^{k+1}, \mathcal{D}^k)\big\|_F \\ &+ \rho(\big\|\mathcal{C}^{k+1} - \mathcal{C}^k\big\|_F + \big\|\mathcal{D}^{k+1} - \mathcal{D}^k\big\|_F). \end{aligned} \tag{35}$$

Since $f(\mathcal{C}, \mathcal{D}) \in C^1$, the partial derivatives $\{\partial_{\mathcal{C}} f(\mathcal{C}, \mathcal{D})\}$ can be assumed to be $L_{\mathcal{C}}$-Lipschitz continuous. Then, we have

$$\big\|\partial_{\mathcal{C}} f(\mathcal{C}^{k+1}, \mathcal{D}^{k+1}) - \partial_{\mathcal{C}} f(\mathcal{C}^{k+1}, \mathcal{D}^k)\big\|_F \leq L_{\mathcal{C}}\big\|\mathcal{D}^{k+1} - \mathcal{D}^k\big\|_F. \tag{36}$$

Substituting (36) into (35) yields

$$\begin{aligned} &\big\|\partial f(\mathcal{C}^{k+1}, \mathcal{D}^{k+1})\big\|_F \\ &\leq L_{\mathcal{C}}\big\|\mathcal{D}^{k+1} - \mathcal{D}^k)\big\|_F \\ &+ \rho\big(\big\|\mathcal{D}^{k+1} - \mathcal{D}^k\big\|_F + \big\|\mathcal{C}^{k+1} - \mathcal{C}^k\big\|_F\big) \\ &\leq (L_{\mathcal{C}} + \rho)\big(\big\|\mathcal{C}^{k+1} - \mathcal{C}^k\big\|_F + \big\|\mathcal{D}^{k+1} - \mathcal{D}^k\big\|_F\big). \end{aligned} \tag{37}$$

Proof of Property 4: The initial point of Algorithm 2, i.e., $(\mathcal{C}^0, \mathcal{D}^0)$, is bounded. Thus, $f(\mathcal{C}^0, \mathcal{D}^0)$ is bounded. Based on the property 2, there has $0 \leq f(\mathcal{C}^k, \mathcal{D}^k) \leq f(\mathcal{C}^0, \mathcal{D}^0)$. Since the continuous function $f(\mathcal{C}, \mathcal{D})$ is proper and coercive, there exists $\|\mathcal{C}\|_F + \|\mathcal{D}\|_F \to \infty$ if and only if $f(\mathcal{C}, \mathcal{D}) \to \infty$. However, $f(\mathcal{C}^k, \mathcal{D}^k)$ is bounded. Then, sequence $\{(\mathcal{C}^k, \mathcal{D}^k)\}_{k\in\mathbb{N}}$ is bounded. □

REFERENCES

[1] Z. Pan, G. Healey, M. Prasad, and B. Tromberg, "Face recognition in hyperspectral images," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 25, no. 12, pp. 1552–1560, Dec. 2003.

[2] R. Senthilkumar, V. Srinidhi, S. Neelavathi, and S. R. Devi, "Face recognition using hyperspectral imaging and deep learning," in *Proc. 10th Int. Conf. Adv. Comput. (ICoAC)*, Dec. 2018, pp. 377–382.

[3] G. Vivone, "Multispectral and hyperspectral image fusion in remote sensing: A survey," *Inf. Fusion*, vol. 89, pp. 405–417, Jan. 2023.

[4] X. Wu, T.-Z. Huang, L.-J. Deng, and T.-J. Zhang, "Dynamic cross feature fusion for remote sensing pansharpening," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 14687–14696.

[5] D. Hong, J. Yao, C. Li, D. Meng, N. Yokoya, and J. Chanussot, "Decoupled-and-coupled networks: Self-supervised hyperspectral image super-resolution with subpixel fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5527812.

[6] J. Xue, Y. Zhao, W. Liao, and J. C. Chan, "Nonlocal low-rank regularized tensor decomposition for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 5174–5189, Jul. 2019.

[7] J. Xue, Y. Zhao, S. Huang, W. Liao, J. C. Chan, and S. G. Kong, "Multilayer sparsity-based tensor decomposition for low-rank tensor completion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6916–6930, Nov. 2022.

[8] J.-L. Xiao, T.-Z. Huang, L.-J. Deng, Z.-C. Wu, and G. Vivone, "A new context-aware details injection fidelity with adaptive coefficients estimation for variational pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5408015.

[9] J.-L. Xiao, T.-Z. Huang, L.-J. Deng, Z.-C. Wu, X. Wu, and G. Vivone, "Variational pansharpening based on coefficient estimation with non-local regression," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5406115.

[10] C. Gomez, R. A. Viscarra Rossel, and A. B. McBratney, "Soil organic carbon prediction by hyperspectral remote sensing and field vis-NIR spectroscopy: An Australian case study," *Geoderma*, vol. 146, nos. 3–4, pp. 403–411, Aug. 2008.

[11] A. Okujeni, S. van der Linden, and P. Hostert, "Extending the vegetation–impervious–soil model using simulated EnMAP data and machine learning," *Remote Sens. Environ.*, vol. 158, pp. 69–80, Mar. 2015.

[12] J. Xue, Y. Zhao, W. Liao, J. C. Chan, and S. G. Kong, "Enhanced sparsity prior model for low-rank tensor completion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 11, pp. 4567–4581, Nov. 2020.

[13] J. Xue, Y. Zhao, Y. Bu, J. C. Chan, and S. G. Kong, "When Laplacian scale mixture meets three-layer transform: A parametric tensor sparsity for tensor completion," *IEEE Trans. Cybern.*, vol. 52, no. 12, pp. 13887–13901, Dec. 2022.

[14] T. Xu, T.-Z. Huang, L.-J. Deng, H.-X. Dou, and N. Yokoya, "TR-STF: A fast and accurate tensor ring decomposition algorithm via defined scaled tri-factorization," *Comput. Appl. Math.*, vol. 42, no. 5, p. 1, Jul. 2023.

[15] C. I. Kanatsoulis, X. Fu, N. D. Sidiropoulos, and W.-K. Ma, "Hyperspectral super-resolution: A coupled tensor factorization approach," *IEEE Trans. Signal Process.*, vol. 66, no. 24, pp. 6503–6517, Dec. 2018.

[16] S. Li, R. Dian, L. Fang, and J. M. Bioucas-Dias, "Fusing hyperspectral and multispectral images via coupled sparse tensor factorization," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 4118–4130, Aug. 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:24:23 UTC from IEEE Xplore. Restrictions apply.

[17] W. He, Y. Chen, N. Yokoya, C. Li, and Q. Zhao, "Hyperspectral super-resolution via coupled tensor ring factorization," *Pattern Recognit.*, vol. 122, Feb. 2022, Art. no. 108280.

[18] N. Yokoya, C. Grohnfeldt, and J. Chanussot, "Hyperspectral and multispectral data fusion: A comparative review of the recent literature," *IEEE Geosci. Remote Sens. Mag. Replaces Newsletter*, vol. 5, no. 2, pp. 29–56, Jun. 2017.

[19] S. Xu, J. Zhang, Z. Zhao, K. Sun, J. Liu, and C. Zhang, "Deep gradient projection networks for pan-sharpening," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1366–1375.

[20] Z. Zhao et al., "Deep convolutional sparse coding networks for interpretable image fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 2368–2376.

[21] Z.-C. Wu, T.-Z. Huang, L.-J. Deng, J.-F. Hu, and G. Vivone, "VO+Net: An adaptive approach using variational optimization and deep learning for panchromatic sharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5401016.

[22] R. Dian, S. Li, and X. Kang, "Regularizing hyperspectral and multispectral image fusion by CNN denoiser," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 3, pp. 1124–1135, Mar. 2021.

[23] J.-F. Hu, T.-Z. Huang, L.-J. Deng, H.-X. Dou, D. Hong, and G. Vivone, "Fusformer: A transformer-based fusion network for hyperspectral image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[24] T.-J. Zhang, L.-J. Deng, T.-Z. Huang, J. Chanussot, and G. Vivone, "A triple-double convolutional neural network for panchromatic sharpening," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 9088–9101, Nov. 2023.

[25] J.-F. Hu, T.-Z. Huang, L.-J. Deng, T.-X. Jiang, G. Vivone, and J. Chanussot, "Hyperspectral image super-resolution via deep spatiospectral attention convolutional neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 7251–7265, Dec. 2022.

[26] R. Dian, S. Li, A. Guo, and L. Fang, "Deep hyperspectral image sharpening," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 11, pp. 5345–5355, Nov. 2018.

[27] D. Shen, J. Liu, Z. Wu, J. Yang, and L. Xiao, "ADMM-HFNet: A matrix decomposition-based deep approach for hyperspectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5513417.

[28] F. Zhou, R. Hang, Q. Liu, and X. Yuan, "Pyramid fully convolutional network for hyperspectral and multispectral image fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 5, pp. 1549–1558, May 2019.

[29] J. Xiao, J. Li, Q. Yuan, and L. Zhang, "A dual-UNet with multistage details injection for hyperspectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5515313.

[30] F. Palsson, J. R. Sveinsson, and M. O. Ulfarsson, "Multispectral and hyperspectral image fusion using a 3-D-convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 639–643, May 2017.

[31] W. Wang, W. Zeng, Y. Huang, X. Ding, and J. Paisley, "Deep blind hyperspectral image fusion," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4149–4158.

[32] Q. Xie, M. Zhou, Q. Zhao, D. Meng, W. Zuo, and Z. Xu, "Multispectral and hyperspectral image fusion by MS/HS fusion net," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 1585–1594.

[33] W. Dong, C. Zhou, F. Wu, J. Wu, G. Shi, and X. Li, "Model-guided deep hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 5754–5768, 2021.

[34] C. Wu, D. Wang, Y. Bai, H. Mao, Y. Li, and Q. Shen, "HSR-Diff: Hyperspectral image super-resolution via conditional diffusion models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 7083–7093.

[35] Z. Cao, S. Cao, L.-J. Deng, X. Wu, J. Hou, and G. Vivone, "Diffusion model with disentangled modulations for sharpening multispectral and hyperspectral images," *Inf. Fusion*, vol. 104, Apr. 2024, Art. no. 102158.

[36] J. Cui, W. Dong, J. Qu, X. Wu, S. Xiao, and Y. Li, "CFMDM: Coarse-to-fine meta-diffusion model for scale-arbitrary hyperspectral super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[37] S. Jia, S. Zhu, Z. Wang, M. Xu, W. Wang, and Y. Guo, "Diffused convolutional neural network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504615.

[38] Y. Xiao, Q. Yuan, K. Jiang, J. He, X. Jin, and L. Zhang, "EDiffSR: An efficient diffusion probabilistic model for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5601514.

[39] L. Zhang, J. Nie, W. Wei, Y. Zhang, S. Liao, and L. Shao, "Unsupervised adaptation learning for hyperspectral imagery super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3073–3082.

[40] J. Li, K. Zheng, W. Liu, Z. Li, H. Yu, and L. Ni, "Model-guided coarse-to-fine fusion network for unsupervised hyperspectral image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[41] L. Gao, J. Li, K. Zheng, and X. Jia, "Enhanced autoencoders with attention-embedded degradation learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509417.

[42] R. Imamura, T. Itasaka, and M. Okuda, "Zero-shot hyperspectral image denoising with separable image prior," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 1416–1420.

[43] Q. Xie, M. Zhou, Q. Zhao, Z. Xu, and D. Meng, "MHF-Net: An interpretable deep network for multispectral and hyperspectral image fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 3, pp. 1457–1473, Mar. 2022.

[44] R. Dian, A. Guo, and S. Li, "Zero-shot hyperspectral sharpening," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 12650–12666, 2023.

[45] R. Kawakami, Y. Matsushita, J. Wright, M. Ben-Ezra, Y.-W. Tai, and K. Ikeuchi, "High-resolution hyperspectral imaging via matrix factorization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2011, pp. 2329–2336.

[46] N. Akhtar, F. Shafait, and A. Mian, "Bayesian sparse representation for hyperspectral image super resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3631–3640.

[47] N. Akhtar, F. Shafait, and A. Mian, "Sparse spatio-spectral representation for hyperspectral image super-resolution," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2014, pp. 63–78.

[48] M. Sim oes, J. Bioucas-Dias, L. B. Almeida, and J. Chanussot, "A convex formulation for hyperspectral image superresolution via subspace-based regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3373–3388, Jun. 2015.

[49] T. Xu, T.-Z. Huang, Y. Chen, J. Huang, and L.-J. Deng, "A variational approach with nonlocal self-similarity and joint-sparsity for hyperspectral image super-resolution," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2021, pp. 2444–2447.

[50] Q. Wei, J. Bioucas-Dias, N. Dobigeon, and J.-Y. Tourneret, "Hyperspectral and multispectral image fusion based on a sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3658–3668, Jul. 2015.

[51] Q. Wei, N. Dobigeon, and J.-Y. Tourneret, "Fast fusion of multi-band images based on solving a Sylvester equation," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4109–4121, Nov. 2015.

[52] J. Zhang, J. Lu, C. Wang, and S. Li, "Hyperspectral and multispectral image fusion via superpixel-based weighted nuclear norm minimization," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5521612.

[53] J. Xue, Y.-Q. Zhao, Y. Bu, W. Liao, J. C.-W. Chan, and W. Philips, "Spatial-spectral structured sparse low-rank representation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 3084–3097, 2021.

[54] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[55] C. Lanaras, E. Baltsavias, and K. Schindler, "Hyperspectral super-resolution by coupled spectral unmixing," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 3586–3594.

[56] E. Wycoff, T.-H. Chan, K. Jia, W.-K. Ma, and Y. Ma, "A non-negative sparse promoting algorithm for high resolution hyperspectral imaging," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, May 2013, pp. 1409–1413.

[57] R. Dian, S. Li, L. Fang, and Q. Wei, "Multispectral and hyperspectral image fusion with spatial-spectral sparse representation," *Inf. Fusion*, vol. 49, pp. 262–270, Sep. 2019.

[58] Y. Xu, Z. Wu, J. Chanussot, and Z. Wei, "Nonlocal patch tensor sparse representation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 28, no. 6, pp. 3034–3047, Jun. 2019.

[59] T. Xu, T.-Z. Huang, L.-J. Deng, and N. Yokoya, "An iterative regularization method based on tensor subspace representation for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5529316.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:24:23 UTC from IEEE Xplore. Restrictions apply.

[60] R. Dian, S. Li, and L. Fang, "Learning a low tensor-train rank representation for hyperspectral image super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 9, pp. 2672–2683, Sep. 2019.
[61] C.-W. Sun, T.-Z. Huang, T. Xu, and L.-J. Deng, "NF-3DLogTNN: An effective hyperspectral and multispectral image fusion method based on nonlocal low-fibered-rank regularization," *Appl. Math. Model.*, vol. 118, pp. 780–797, Jun. 2023.
[62] Y. Xu, Z. Wu, J. Chanussot, P. Comon, and Z. Wei, "Nonlocal coupled tensor CP decomposition for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 348–362, Jan. 2020.
[63] T. Xu, T.-Z. Huang, L.-J. Deng, X.-L. Zhao, and J. Huang, "Hyperspectral image superresolution using unidirectional total variation with tucker decomposition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4381–4398, 2020.
[64] Y. Chen, J. Zeng, W. He, X.-L. Zhao, and T.-Z. Huang, "Hyperspectral and multispectral image fusion using factor smoothed tensor ring decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5515417.
[65] T. G. Kolda and B. W. Bader, "Tensor decompositions and applications," *SIAM Rev. Soc. Ind. Appl. Math.*, vol. 51, no. 3, pp. 455–500, Aug. 2009.
[66] Z.-C. Wu, T.-Z. Huang, L.-J. Deng, H.-X. Dou, and D. Meng, "Tensor wheel decomposition and its tensor completion application," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, p. 5.
[67] Q. Zhao, G. Zhou, S. Xie, L. Zhang, and A. Cichocki, "Tensor ring decomposition," 2016, *arXiv:1606.05535*.
[68] H. Attouch, J. Bolte, and B. F. Svaiter, "Tensor decompositions and applicatioconvergence of descent methods for semi-algebraic and tame problems: Proximal algorithms, forward–backward splitting, and regularized Gauss–Seidel methods," *Math. Program.*, vol. 137, pp. 91–129, Jan. 2013.
[69] H. Attouch, J. Bolte, P. Redont, and A. Soubeyran, "Proximal alternating minimization and projection methods for nonconvex problems: An approach based on the Kurdyka–Łojasiewicz inequality," *Math. Oper. Res.*, vol. 35, no. 2, pp. 438–457, May 2010.
[70] G. H. Golub and C. F. Van Loan, *Matrix Computations*. Baltimore, MD, USA: The Johns Hopkins Univ. Press, 2012.
[71] N. Yokoya and A. Iwasaki, "Airborne hyperspectral data over Chikusei," Space Appl. Lab., Univ. Tokyo, Tokyo, Japan, Tech. Rep., SAL-2016-05-27, May 2016.
[72] W. Zhou, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, pp. 600–612, 2004.
[73] L. Alparone, L. Wald, J. Chanussot, C. Thomas, P. Gamba, and L. M. Bruce, "Comparison of pansharpening algorithms: Outcome of the 2006 GRS-S data-fusion contest," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3012–3021, Oct. 2007.
[74] A. F. G. J. R. H. Yuhas and J. M. Boardman, "Discrimination among semi-arid landscape endmembers using the spectral angle mapper (SAM) algorithm," in *Proc. Summaries Annu. JPL Airborne Geosci. Workshop*, 1992, pp. 147–149.
[75] J. S. Pearlman, P. S. Barry, C. C. Segal, J. Shepanski, D. Beiso, and S. L. Carman, "Hyperion, a space-based imaging spectrometer," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 6, pp. 1160–1173, Jun. 2003.
[76] J. Bolte, S. Sabach, and M. Teboulle, "Proximal alternating linearized minimization for nonconvex and nonsmooth problems," *Math. Program.*, vol. 146, pp. 459–494, Aug. 2014.
[77] H. Attouch, J. Bolte, and B. F. Svaiter, "Convergence of descent methods for semi-algebraic and tame problems: Proximal algorithms, forward–backward splitting, and regularized Gauss–Seidel methods," *Math. Program.*, vol. 137, nos. 1–2, pp. 91–129, Feb. 2013.



Ting Xu received the B.S. degree in mathematics and applied mathematics from the Mianyang Teachers' College, Mianyang, China, in 2018. She is currently pursuing the Ph.D. degree in mathematics with the School of Mathematical Sciences, University of Electronic Science and Technology of China (UESTC), Chengdu, China.

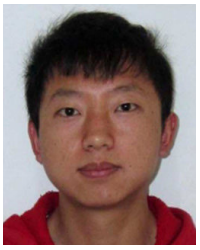
From 2023 to 2024, she was an Intern with the RIKEN Center for Advanced Intelligence Project, the Geoinformatics Team, Tokyo, Japan. Her research interests include image processing, image fusion, tensor decomposition, and deep learning.



Ting-Zhu Huang (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in computational mathematics from the Department of Mathematics, Xi'an Jiaotong University, Xi'an, China, in 1986, 1992, and 2001, respectively.

He is currently a Professor with the School of Mathematical Sciences, University of Electronic Science and Technology of China, Chengdu, China. His research interests include scientific computation and applications, numerical algorithms for image processing, numerical linear algebra, preconditioning technologies, and matrix analysis with applications.

Dr. Huang is an Editor of *The Scientific World Journal*, *Advances in Numerical Analysis*, the *Journal of Applied Mathematics*, the *Journal of Pure and Applied Mathematics: Advances in Applied Mathematics*, and the *Journal of Electronic Science and Technology*, China.



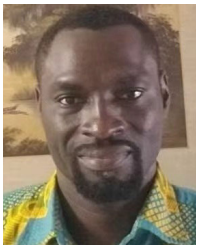
Liang-Jian Deng (Senior Member, IEEE) received the B.S. and Ph.D. degrees in applied mathematics from the School of Mathematical Sciences, University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2010 and 2016, respectively.

From 2013 to 2014, he was a Joint-Training Ph.D. Student with the Case Western Reserve University, Cleveland, OH, USA. In 2017, he was a Post-Doctoral Researcher with Hong Kong Baptist University (HKBU), Hong Kong. In addition, he also stayed at the Isaac Newton Institute for Mathematical Sciences, Cambridge University, Cambridge, U.K., and HKBU for short visits. He is currently a Professor with the School of Mathematical Sciences, UESTC. His research interests include the use of partial differential equations (PDEs), optimization modeling, and deep learning to address several tasks in image processing and computer vision, e.g., resolution enhancement and restoration.



Jin-Liang Xiao (Graduate Student Member, IEEE) received the B.S. degree in mathematics from the School of Mathematics and Statistics, Yunnan University (YNU), Kunming, China, in 2019. He is currently pursuing the Ph.D. degree with the School of Mathematical Sciences, University of Electronic Science and Technology of China, Chengdu, China.

From 2016 to 2018, he was a Joint-Training B.S. Student with the School of Mathematical Sciences, Fudan University, Shanghai, China. Since November 2023, he has been an Intern with the RIKEN Center for Advanced Intelligence Project, Tensor Learning Team, Tokyo, Japan. His research interests include image fusion and low-rank image processing.



Clifford Broni-Bediako (Member, IEEE) received the B.Sc. degree in computer science from the Kwame Nkrumah University of Science and Technology, Kumasi, Ghana, in 2003, the M.Phil. degree in computer science from the University of Ghana, Accra, Ghana, in 2018, and the Ph.D. degree in engineering from Soka University, Tokyo, Japan, in 2022.

Since 2022, he has been a Post-Doctoral Researcher with the RIKEN Center for Advanced Intelligence Project (AIP), Geoinformatics Team, Tokyo, Japan. His research revolves around deep learning, particularly efficient deep neural networks, automated neural architecture search, and remote sensing image understanding with deep learning algorithms.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:24:23 UTC from IEEE Xplore. Restrictions apply.



Junshi Xia (Senior Member, IEEE) received the B.S. degree in geographic information systems and the Ph.D. degree in photogrammetry and remote sensing from the China University of Mining and Technology, Xuzhou, China, in 2008 and 2013, respectively, and the Ph.D. degree in image processing from the Grenoble Images Speech Signals and Automatics Laboratory, Grenoble Institute of Technology, Grenoble, France, in 2014.

From 2014 to 2015, he was a Visiting Scientist with the Department of Geographic Information Sciences, Nanjing University, Nanjing, China. From 2015 to 2016, he was a Post-Doctoral Research Fellow with the University of Bordeaux, Bordeaux, France. From 2016 to 2018, he was the Japan Society for the Promotion of Science (JSPS) Post-Doctoral Overseas Research Fellow with The University of Tokyo, Tokyo, Japan. Since 2018, he has been with the RIKEN Center for Advanced Intelligence Project (AIP), Tokyo, where he is currently a Senior Research Scientist. His research interests include multiple classifier systems in remote sensing, hyperspectral remote sensing image processing, and deep learning in remote sensing applications.

Dr. Xia was a recipient of the First Place Prize in the IEEE Geoscience and Remote Sensing Society Data Fusion Contest organized by the Image Analysis and Data Fusion Technical Committee in 2017. Since 2019, he has been an Associate Editor of IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, *Remote Sensing*, and *Frontiers in Remote Sensing*, and a Guest Editor of *Remote Sensing* and IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS).



Naoto Yokoya (Member, IEEE) received the M.Eng. and Ph.D. degrees from the Department of Aeronautics and Astronautics, The University of Tokyo, Tokyo, Japan, in 2010 and 2013, respectively.

He is currently an Associate Professor at The University of Tokyo and a Team Leader at the RIKEN Center for Advanced Intelligence Project, Tokyo, where he leads the Geoinformatics Team. He was an Assistant Professor at the University of Tokyo from 2013 to 2017. From 2015 to 2017, he was an Alexander von Humboldt Fellow, working at the German Aerospace Center (DLR), Oberpfaffenhofen, Munich, Germany, and the Technical University of Munich (TUM), Munich. His research interests include the development of image processing, data fusion, and machine learning algorithms for understanding remote sensing images, with applications to disaster management.

Dr. Yokoya won First Place in the 2017 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest organized by the Image Analysis and Data Fusion Technical Committee (IADF TC). He was the Chair (2019–2021) and a Co-Chair (2017–2019) of IEEE GRSS IADF TC. He has been an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING since 2021 and the *ISPRS Journal of Photogrammetry and Remote Sensing* since 2024. He has been designated a Clarivate Highly Cited Researcher since 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 18:24:23 UTC from IEEE Xplore. Restrictions apply.